

AuroraPhoto

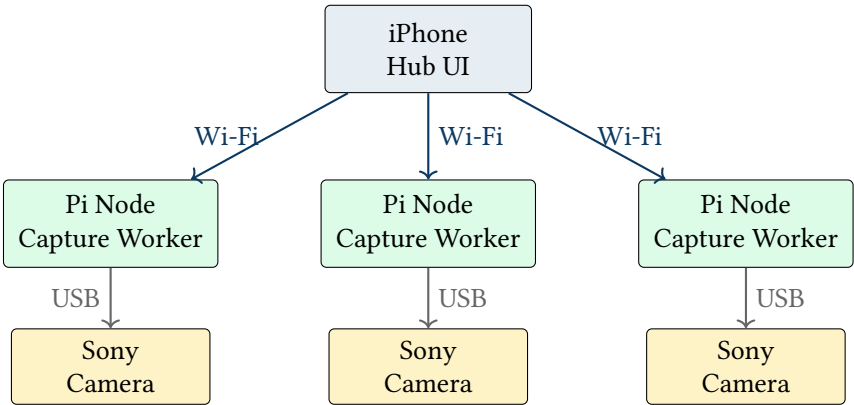
Distributed Edge Control for Adaptive Aurora Capture and Star Sharpness

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AuroraPhoto Project

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Abstract. AuroraPhoto is a distributed astrophotography control system aimed at capturing high-quality aurora and night-sky imagery under rapidly changing conditions. The core design uses an iPhone as a field-ready coordination hub and one or more Raspberry Pi capture nodes connected to Sony Alpha cameras over USB. Each node performs local (edge) image analysis to compute star sharpness (Half-Flux Radius, HFR) and exposure histograms from preview imagery, enabling low-latency focus correction and adaptive exposure profile switching during aurora “bursts”. This report specifies the target architecture, key control loops, failure modes specific to cold-field USB operation, and a roadmap-to-requirements traceability model suitable for implementation and validation.

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Acronyms and Terms

Term	Meaning
HFR	Half-Flux Radius; a star sharpness proxy computed from extracted point sources. Lower is sharper.
SNR	Signal-to-noise ratio; proxy for whether stars can be reliably measured in previews.
p95/p99	95th/99th percentile of a latency distribution; used to reason about tail latency.
Delta-trigger	Trigger scheduling method where nodes receive a command to act at a future timestamp to absorb network jitter.

Table 1. Acronyms and terms used in this report.

1 Problem Framing

Aurora photography presents an unusually adversarial control environment: scene illumination changes rapidly; the most visually valuable structure (discrete rays and curtains) can require short exposures; focus drifts with temperature; and field constraints (cold, battery, unreliable links) magnify small failure modes.

Design Decision 1: Edge-first Control

The system performs all time-critical perception (histogram checks, star sharpness) on the Raspberry Pi node. The phone coordinates, visualizes, and triggers, but does not become a compute bottleneck.

Insight:
The key product value is not “automation” as in “unattended intervalometer capture; it is closed-loop control where preview analysis feeds back into exposure and focus decisions within seconds.

2 System Model and Component Boundaries

We model AuroraPhoto as a hub-and-spoke distributed control system consisting of three primary roles: a mobile coordination hub, one or more edge capture nodes, and camera devices.

2.1 Component Responsibilities

Component	Primary Responsibility	Interface
iPhone Hub	UI, settings, composition aids (compass/overlay), coordinated triggering, status dashboard	Wi-Fi
Pi Capture Node	Camera control, preview analysis (HFR, histogram), local buffering, watchdog/recovery	USB + Wi-Fi
Sony Camera	RAW acquisition, focus motor control, exposure parameter execution	USB (PTP/SDK)

Table 2. AuroraPhoto component boundaries and responsibilities.

2.2 Interfaces and Constraints

Two constraints dominate the architecture:

1. Preview-to-decision latency must be low (seconds) to preserve aurora beam definition.
2. Bandwidth is scarce in the field; RAW transfer is prohibitively expensive for real-time loops.

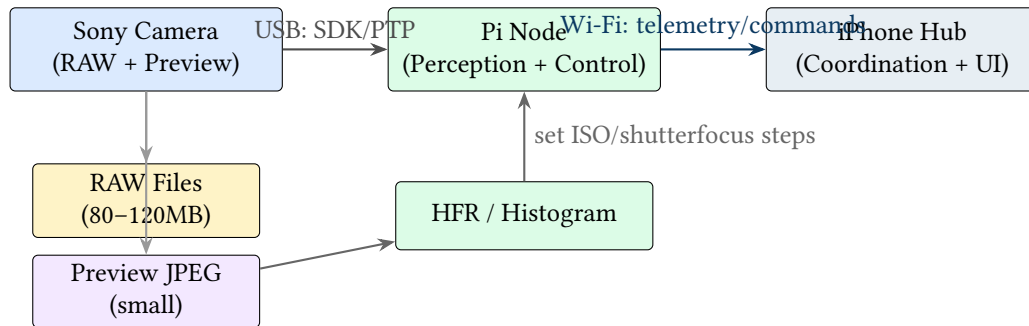


Figure 1. Data movement is asymmetric: previews drive decisions; RAW is buffered and transferred opportunistically.

3 Control Loops

AuroraPhoto is best described as three interacting control loops running at different time scales.

3.1 Coupled-loop View (Focus, Exposure, Burst)

The loops are coupled through shared observables (preview frames) and shared actuators (camera parameters). A useful mental model is a triad of controllers operating on different objectives but competing for the same camera time.

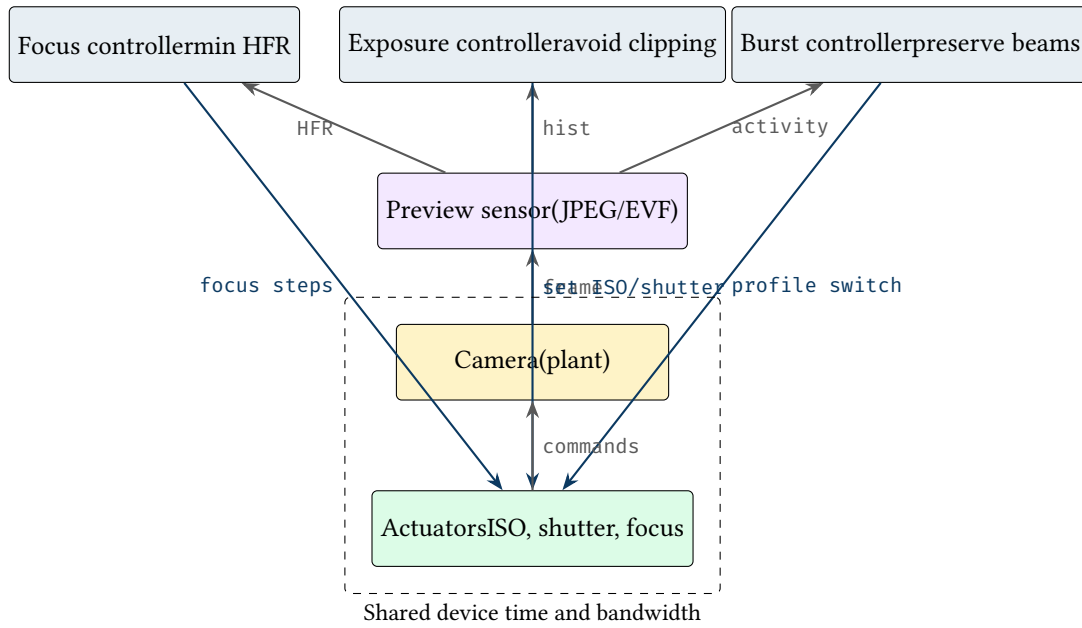


Figure 2. Coupled-loop view: three controllers observe the same preview stream and compete for shared actuation bandwidth.

3.2 Focus Loop (HFR Minimization)

The focus loop uses preview frames to detect stars and compute a scalar sharpness metric (HFR). The control objective is to minimize HFR by stepping the focus motor.

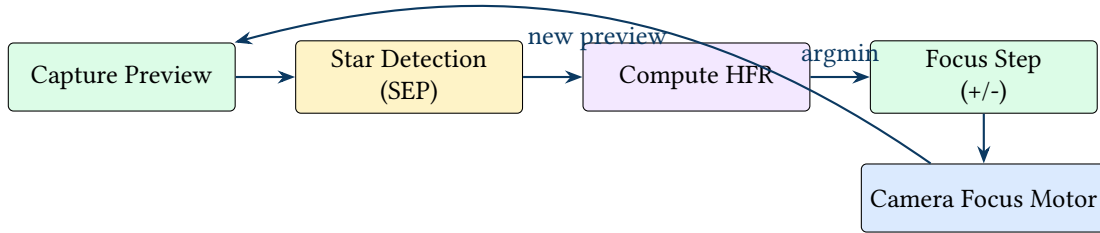


Figure 3. Focus loop: preview-driven HFR minimization on the edge node.

3.3 Exposure Loop (Histogram Guard Rails)

Aurora scenes often clip the green channel; once clipped, beam structure is irrecoverable.

: Green-channel Clip Avoidance

Treat preview histogram clipping as a hard constraint: if any channel exceeds a clipping threshold, shorten shutter and/or reduce ISO before continuing the sequence.

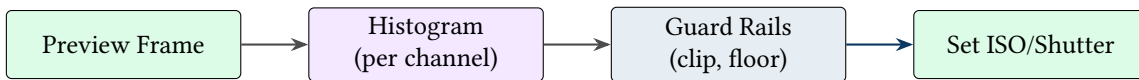


Figure 4. Exposure loop: histogram-based constraint enforcement to protect highlight detail.

3.4 Burst Loop (Activity-driven Profile Switching)

Burst logic switches between exposure profiles (e.g., “wait” vs “burst”) based on detected aurora activity. In v1, a heuristic can be histogram-based; in v2, ML classification is targeted.

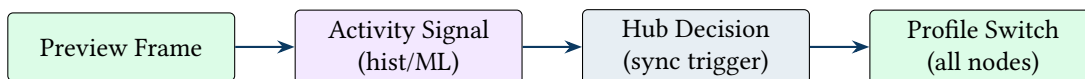


Figure 5. Burst loop: activity detection escalates capture aggressiveness while preserving synchronization.

4 Distributed Synchronization

Multi-node capture requires synchronized triggering while tolerating intermittent connectivity.

Design Decision 2: Delta-triggering

Nodes maintain a shared schedule but wait for a “delta” trigger message from the hub. This reduces sensitivity to clock skew while enabling aligned bursts.

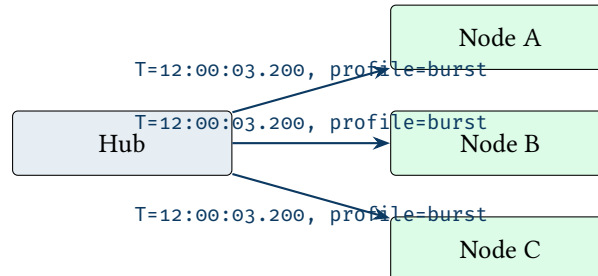


Figure 6. Delta-triggering: a hub message specifies a future trigger timestamp to absorb network jitter.

5 Reliability and Field Failure Modes

Field conditions expose failure classes not common in lab camera automation.

5.1 Sony USB “Hang” and Recovery

Sony cameras can appear present on the USB bus yet be unresponsive to PTP commands. The recovery strategy is to detect timeouts and power-cycle the relevant USB port (via a hub with per-port power switching).

USB Hang Risk

If recovery requires a physical unplug/replug, the system ceases to be “remote”. USB reset via controllable hubs is a design requirement (SHRP-04).

5.2 Thermal Focus Drift

Even if focus is correct at startup, temperature-driven drift can invalidate the imaging run. Monitoring HFR provides a quantitative trigger for refocus routines.

6 Requirements Traceability

7 End-to-End Operational Scenarios

This section describes the primary user-visible workflows as message-and-control sequences. The intent is to make the architecture testable: each scenario implies timing expectations, failure handling, and a minimum set of state transitions.

7.1 Scenario A: Field Bring-up and Node Enrollment

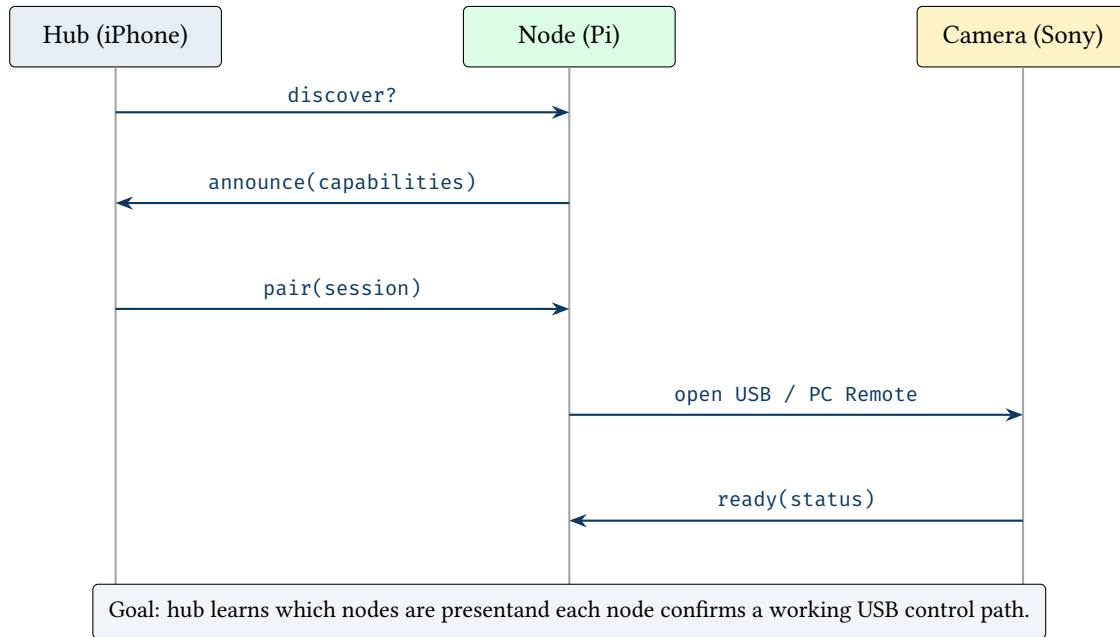


Figure 7. Sequence: bring-up and enrollment (conceptual protocol).

7.2 Scenario B: Focus Calibration at Dusk

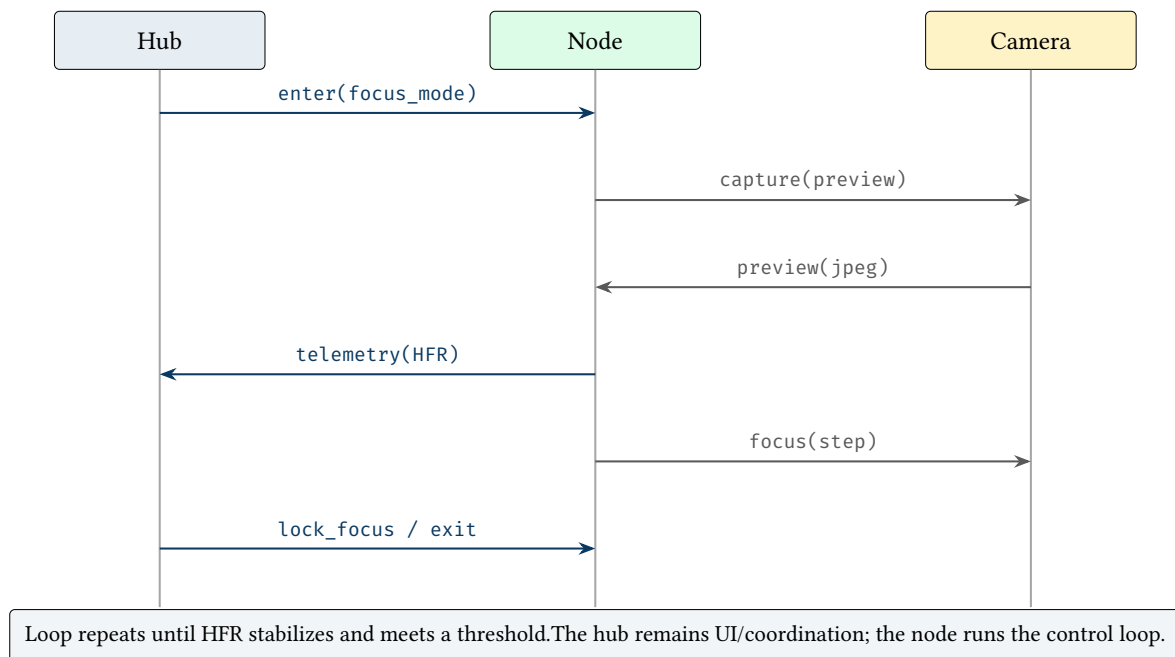


Figure 8. Sequence: focus calibration loop at dusk (conceptual).

7.3 Scenario C: Aurora Burst with Multi-node Synchronization

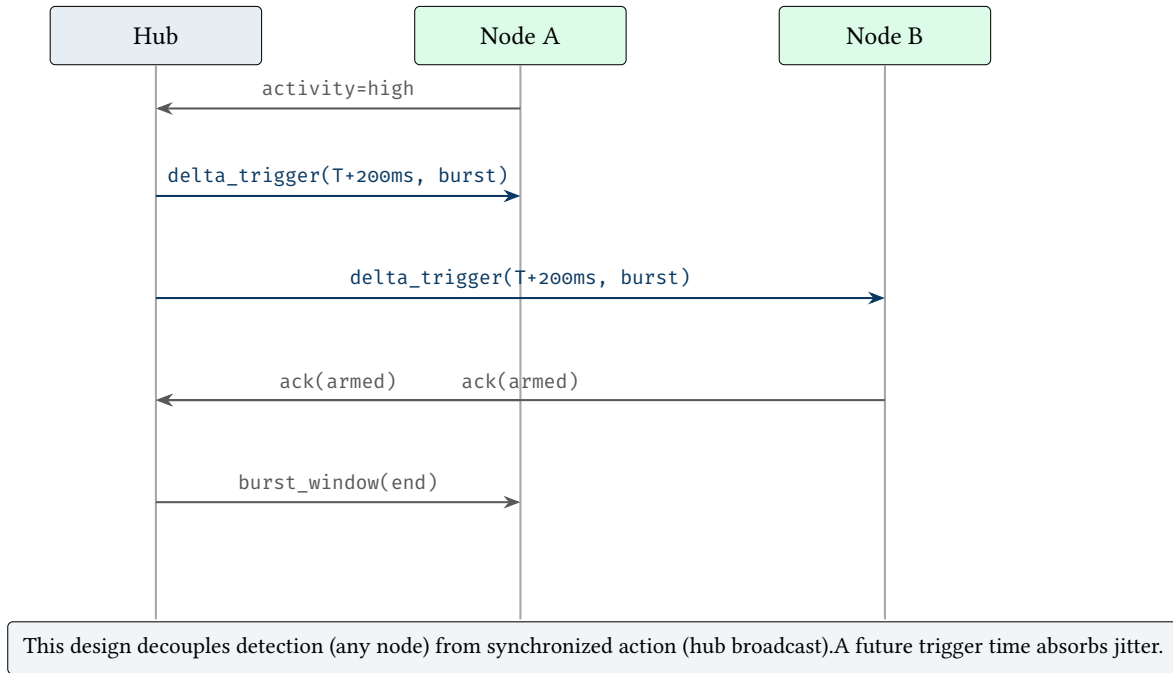


Figure 9. Sequence: one node detects activity; hub coordinates a synchronized burst.

8 Formal State Models

The system is safety-critical in the sense of opportunity cost: failure wastes an entire night of data. Clear state models simplify implementation and test harness design.

8.1 Capture Node State Machine

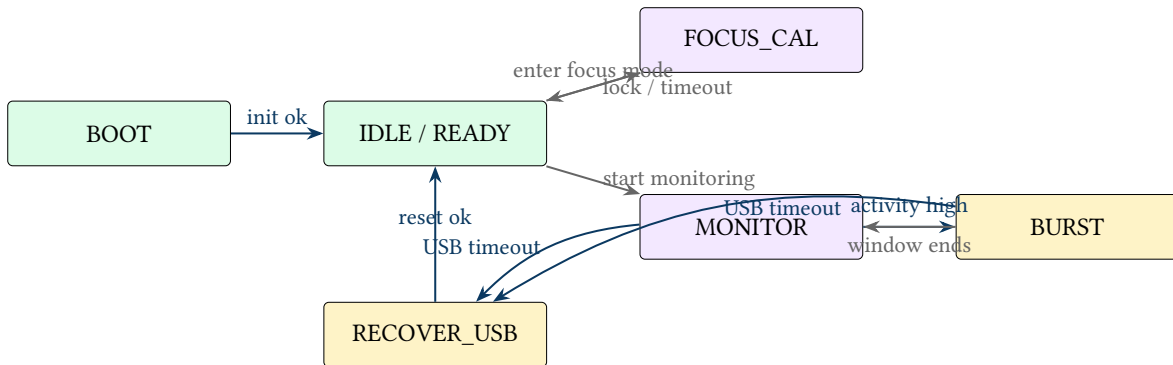


Figure 10. Conceptual node state machine including USB recovery as a first-class state.

8.2 Hub State Machine

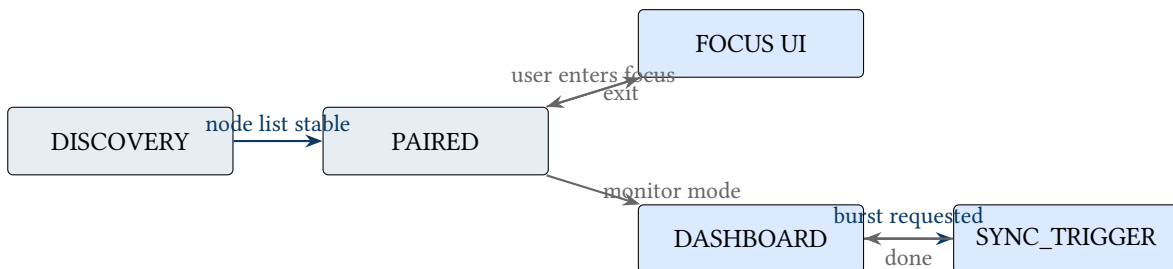


Figure 11. Conceptual hub state machine centered on coordination, not computation.

9 Timing and Performance Budgets

Although this report is conceptual, the control loops imply quantitative budgets. The key metric is closed-loop latency: the time from a preview frame being captured to a control action being applied.

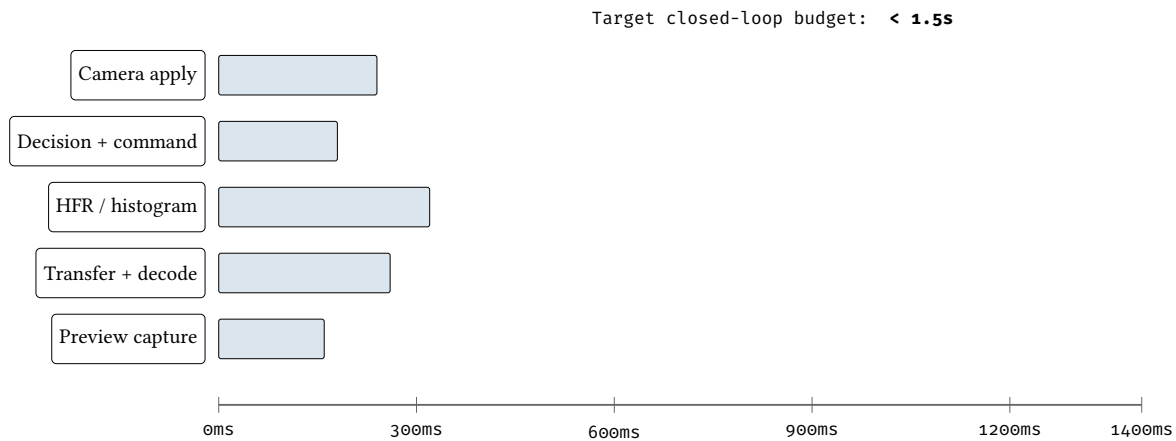


Figure 12. Conceptual latency budget for a single control iteration (focus/exposure).

9.1 Algorithmic Detail: HFR Estimation Pipeline (Per-frame)

HFR estimation is the core “advanced” perception primitive. The algorithm below is intentionally framed as a pipeline with uncertainty gates rather than a linear flowchart: each stage can emit confidence and failure reasons.

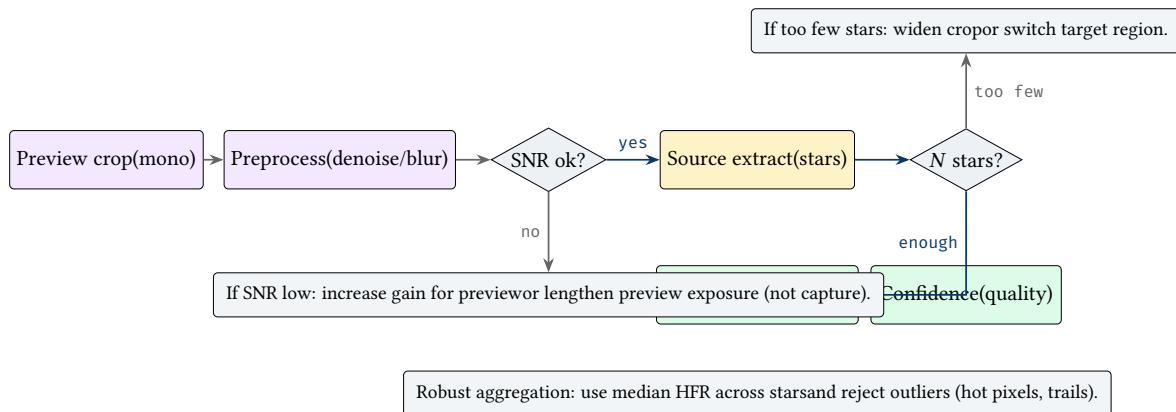


Figure 13. HFR estimation as a gated perception pipeline with explicit failure modes and confidence.

9.2 Algorithmic Detail: Burst Detection as a Change-point Problem

“Burst” events can be framed as change points in a time series of preview-derived signals (brightness, texture, motion, or ML scores). This motivates designs that use hysteresis, smoothing, and minimum-duration windows.

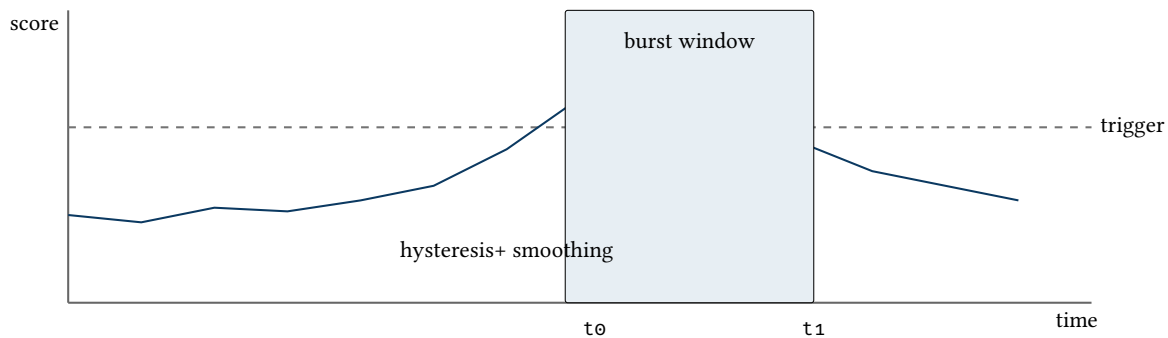


Figure 14. Burst detection as change-point detection on a scalar activity score with hysteresis and minimum-duration windows.

9.3 Algorithmic Detail: Focus Search Under Noise (Coarse-to-fine)

Autofocus for stars is an optimization under noisy measurements: each focus step yields a new preview frame and a stochastic HFR estimate. A robust strategy is a coarse-to-fine search: bracket the minimum with large steps, then refine with smaller steps and a stop condition based on improvement and measurement confidence.

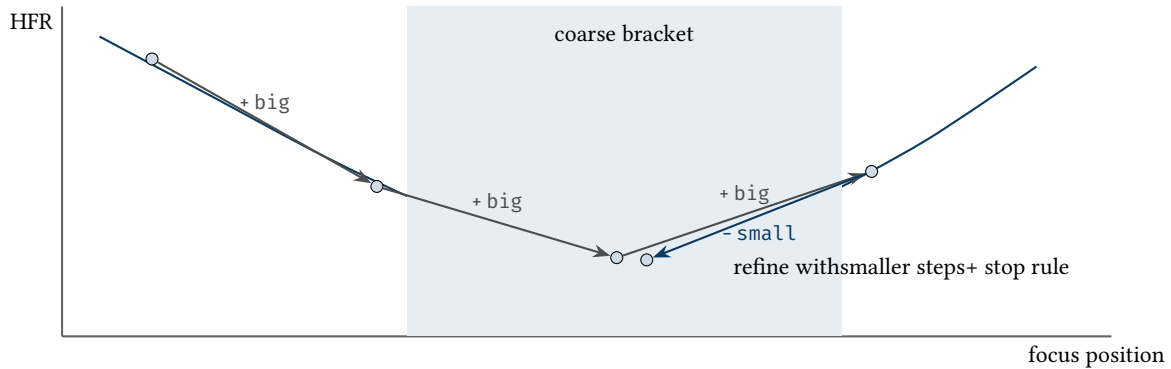


Figure 15. Focus search under noise: bracket the minimum with large steps, then refine locally with smaller steps and confidence-aware stopping.

9.4 Algorithmic Detail: Exposure Trade-offs as a Pareto Frontier

Aurora capture optimizes multiple competing objectives: avoid star trailing (short shutter), preserve beam structure (short shutter), avoid clipping (lower ISO/shutter), and avoid excessive noise (lower ISO / sufficient exposure). These are not simultaneously satisfiable; the system should operate on a Pareto frontier and switch regimes during bursts.

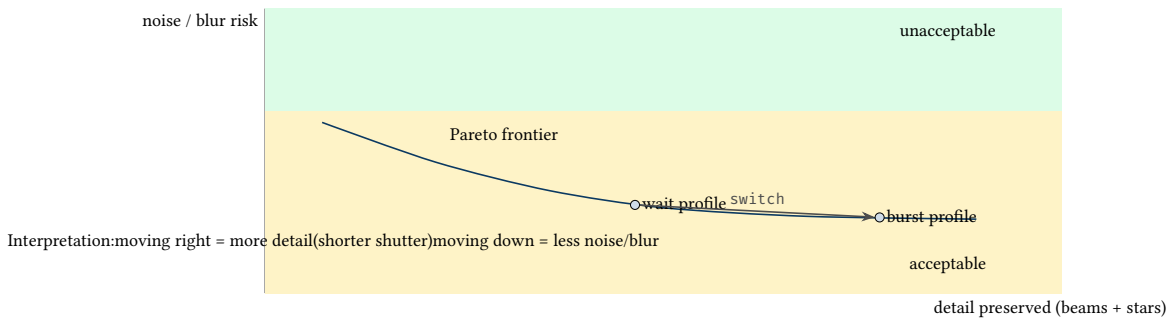
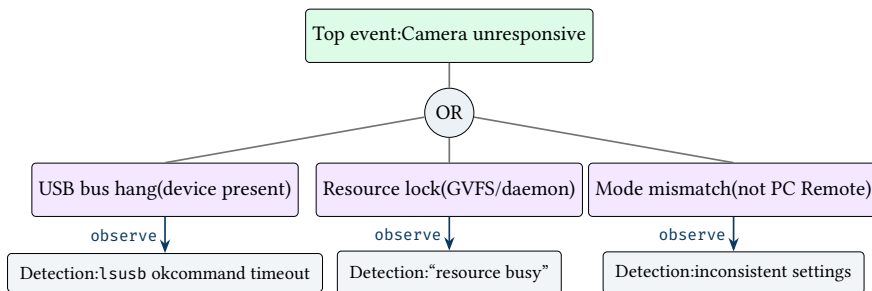


Figure 16. Multi-objective exposure tuning: operate along a Pareto frontier and switch operating points when activity changes.

9.5 Reliability Analysis: Fault Tree and Recovery Confidence

Many failures manifest identically (“no response”) but have different remedies. A lightweight fault tree helps separate detection signals from actionability and allows the system to emit a recovery confidence value.



Recovery actions: (1) power-cycle USB port (2) kill locking process (3) set PC Remote + reconnect. Emit: **recovery confidence** based on which leaf fired.

Figure 17. Fault tree: separate failure causes that share the same symptom, and tie each cause to a specific recovery action and confidence.

10 Design Rules and Anti-patterns (Field-driven)

This section elevates domain pitfalls into explicit design constraints.

: No RAW in Feedback Loops

Do not rely on RAW transfer for any real-time decision. Preview-derived metrics are the only viable feedback signal in the field.

: Recovery is a State

USB hang recovery is not an exception path; it is an expected state transition with explicit detection, actuation (power-cycle), and a return-to-service criterion.

: Edge-heavy Perception

All perception that closes a loop (HFR, histogram, activity) executes on the node to avoid Wi-Fi latency and bandwidth coupling.

11 Implementation Roadmap (Conceptual)

Even without committing to a specific API stack, the work packages can be defined by which loop they enable and how they can be validated.

Phase	Primary Deliverable	Validation
P1	Node USB control + preview acquisition	Repeatable shutter/ISO/focus commands in cold soak
P2	HFR and histogram metrics	HFR correlates with visible star sharpness; no highlight clipping
P3	Hub UI + telemetry dashboard	Field setup in gloves; stable node status and simple triggers
P4	Burst profile switching + delta-triggering	Multi-node burst alignment within tolerance; minimal missed bursts

Table 3. Conceptual roadmap aligned to control loops and validation.

12 Assumptions, Invariants, and Testable Claims

This report is intentionally conceptual (no fixed API), but it still aims to be falsifiable. Each claim below implies a test harness and measurable acceptance criteria.

12.1 System Assumptions

1. A low-resolution preview stream is available at a cadence sufficient for control (order 1–5 Hz).
2. Focus steps are repeatable (monotonic mapping between command and focus motor motion).
3. Preview-derived metrics (HFR, histogram, activity score) correlate strongly with the desired outcomes (sharp stars, unclipped highlights, preserved beam structure).

12.2 Invariants

: Safety over Aggression

If the histogram indicates clipping risk, the system must de-rate exposure even if that reduces activity score sensitivity.

: Loop Independence via Priority

When control loops conflict, resolve by priority: (1) prevent clipping, (2) maintain focus, (3) optimize burst/detail.

12.3 Testable Claims

Claim	Statement	How to Test
C1	Closed-loop exposure adjustments occur within a bounded latency window under nominal conditions.	Inject step-change brightness and measure time-to-command.
C2	HFR estimates are stable enough to drive focus search without oscillation.	Run repeated focus sweeps and compute variance / convergence.
C3	Burst switching preserves beam detail compared to a fixed-profile baseline.	A/B capture sequences; compare clipped pixels and structure metrics.
C4	USB hang recovery returns the node to service without human intervention in most cases.	Cold soak + scripted fault injection; count successful recoveries.

Table 4. Representative claims and validation strategies.

13 Validation Protocols

This section proposes field-realistic protocols. The goal is not perfect lab repeatability; it is a disciplined way to learn which assumptions fail.

13.1 Protocol: Preview Cadence and Latency Budget

Measure the end-to-end time from preview capture to actuation. Record distributions (median, 95th percentile), not just means.

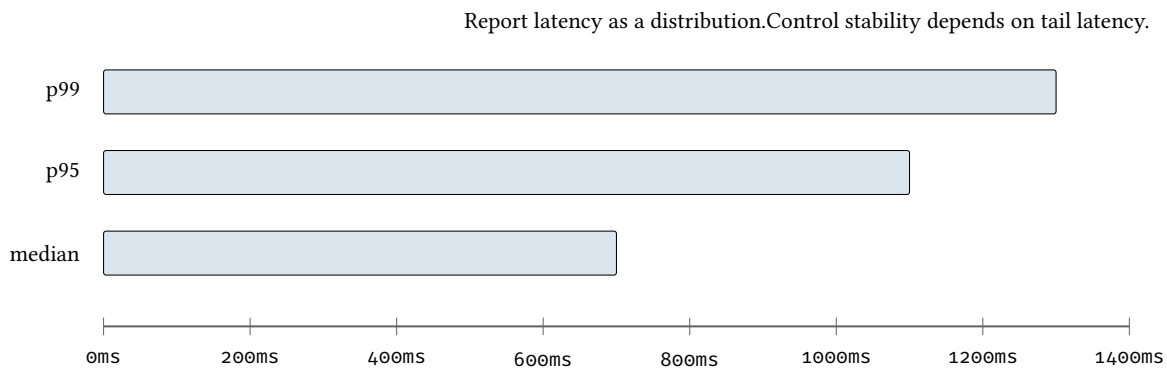


Figure 18. Latency reporting template (illustrative).

13.2 Protocol: Focus Drift and Refocus Triggers

Track HFR over time while temperature drops. Identify a drift threshold that predicts visible star degradation, then convert it into an alert or refocus trigger.

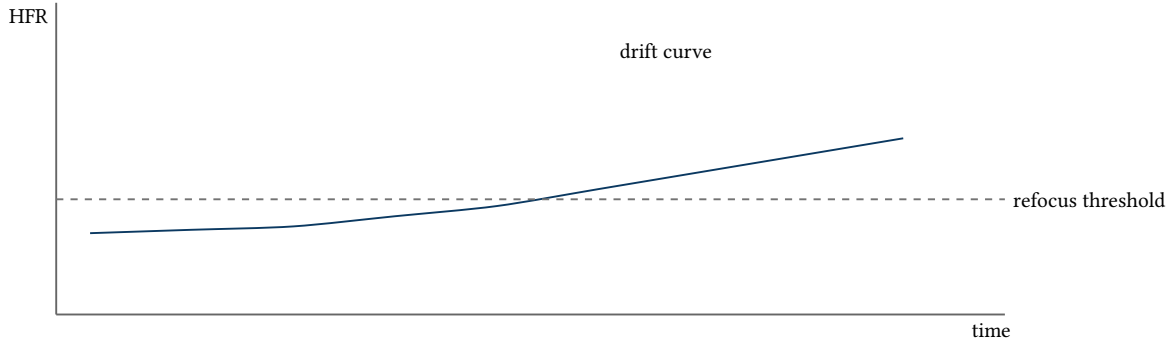


Figure 19. Focus drift protocol: estimate the threshold where refocus becomes necessary (illustrative).

13.3 Protocol: Burst Benefit A/B Test

Capture comparable sequences with (A) fixed exposure and (B) burst switching. Compare clipped pixels, median star sharpness, and a proxy for aurora structure preservation.

14 Open Questions and Research Risks

The following items are deliberately left as open research questions because their answers depend on real field performance.

1. Preview stream cadence/latency on target camera models under cold conditions.
2. The robustness of HFR estimates when aurora brightness overwhelms star contrast.
3. The best activity signal for burst switching in v1 (histogram heuristics vs texture/motion vs ML).
4. Multi-node synchronization tolerance required for downstream use (stitching vs simple multi-angle capture).

15 Estimation and Control Under Uncertainty

This section introduces a unifying mental model: the system maintains an internal belief about the scene and device health, then uses that belief to choose actions. The belief is not a full probabilistic model; it is a compact set of scores (HFR, histogram clip margin, activity score, device health) paired with confidence values.

15.1 Belief State and Confidence Gates

Design Decision 3: Confidence-first Control

The system should not act aggressively on low-confidence estimates. Confidence gates convert uncertain perception into conservative actions or requests for more evidence (more frames, wider crop, different preview settings).

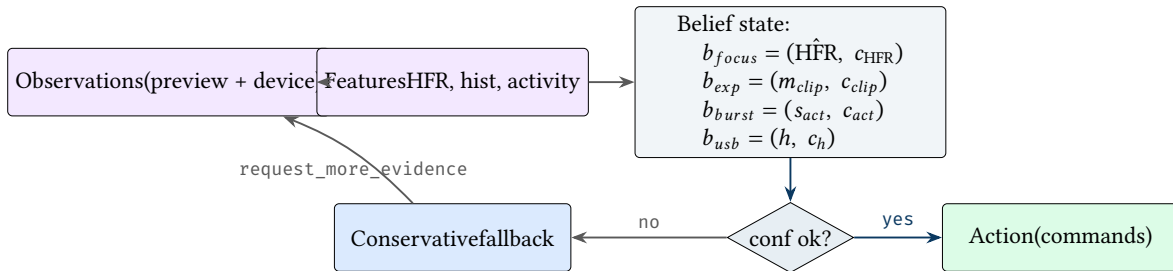


Figure 20. Belief-state framing: estimates are paired with confidence, and confidence gates determine whether to act or gather more evidence.

15.2 Uncertainty Propagation (Conceptual)

To avoid brittle behavior, confidence should decrease when conditions degrade (e.g., aurora overwhelms star contrast). The system can treat confidence as a function of SNR and star count.

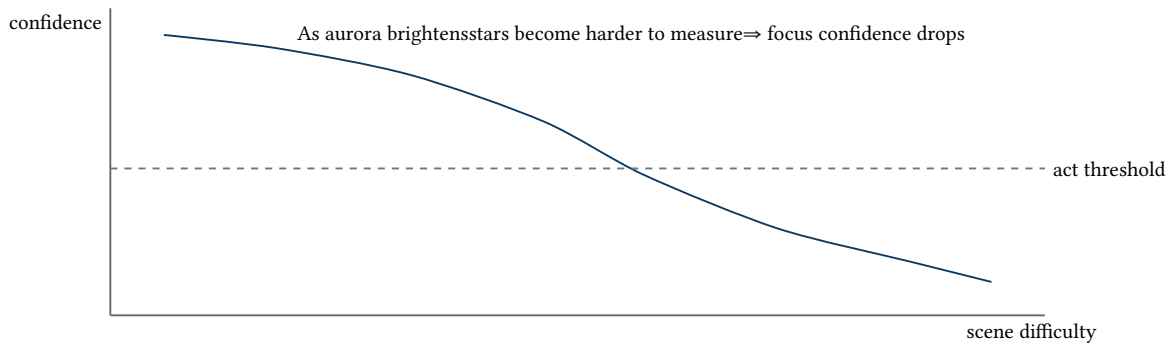


Figure 21. Confidence decreases with increasing scene difficulty; controllers de-rate actions below a threshold.

16 Test Harness Design

This section defines how to validate the system without relying on perfect aurora occurrences. The approach is to separate perception validation (offline datasets), control validation (synthetic perturbations), and reliability validation (fault injection).

16.1 Experiment Matrix (Fault Injection)

Fault	Detect	Recover	Assert	Score
USB hang	timeout	power-cycle	ready again	p_{rec}
Resource lock	busy	kill daemon	connect ok	p_{rec}
Low SNR	gate fail	needs tuning	safe mode	c

Key: treat each row as an executable test case. p_{rec} : recovery success probability estimate; c : confidence calibration.

Figure 22. Fault injection matrix template: each row defines a reproducible test with detection, recovery, and assertions.

16.2 Calibration: Recovery Confidence

Recovery confidence should be calibrated empirically. After repeated injections, the system can map detected leaf causes to observed recovery rates.

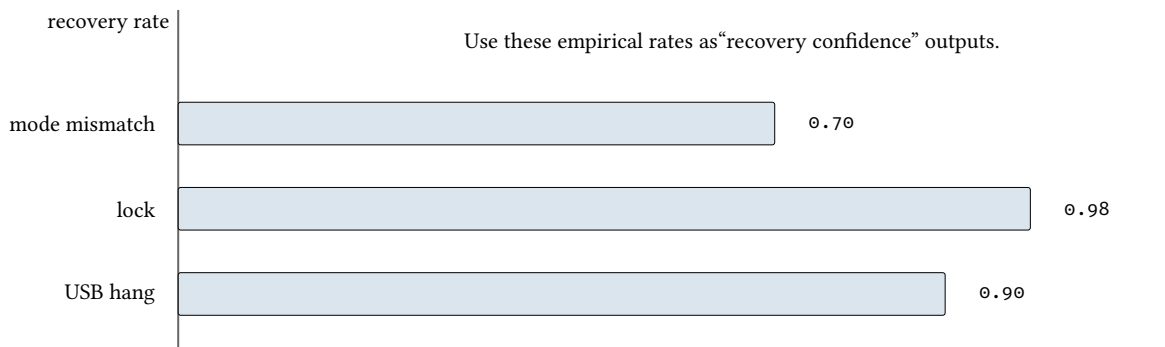


Figure 23. Recovery confidence calibration (illustrative): map detected fault classes to empirical recovery rates.

17 Multi-node Theory and Synchronization Error Budget

Multi-node capture is governed by the distribution of trigger misalignment. Misalignment is the sum of multiple independent (or partially correlated) sources.

17.1 Error Decomposition

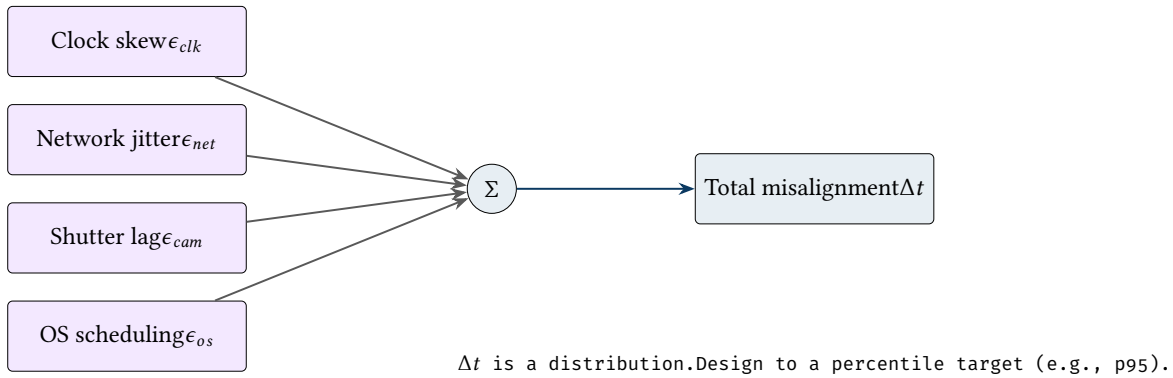


Figure 24. Synchronization error budget: decompose misalignment into clock, network, camera, and OS components.

17.2 Timing Diagram: Delta-trigger vs Immediate Trigger

Delta-triggering reduces sensitivity to network jitter by scheduling a future time.

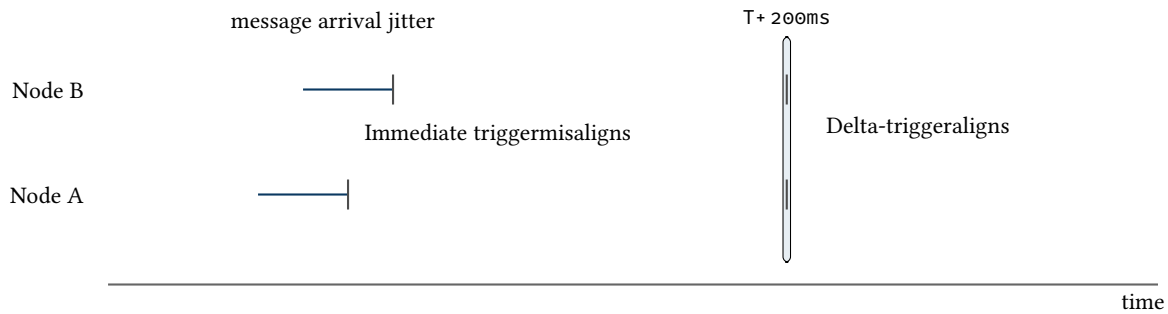


Figure 25. Delta-triggering aligns actions at a future timestamp despite message arrival jitter.

17.3 Tolerance Targets

Tolerance should be specified by downstream goals. For example, panorama stitching may tolerate looser sync than any attempt at parallax-based effects.

Use Case	Target (p95)	Notes
Multi-angle documentation	250 ms	Human perception tolerant; goal is capture coverage.
Panorama stitching (static scenes)	100 ms	Aurora motion may still dominate; depends on focal length.
Synchronized burst aesthetics	50 ms	More sensitive; requires tighter clocks and predictable shutter lag.

Table 5. Illustrative synchronization tolerance targets (to be validated).

The following requirements identifiers define v1 scope and support roadmap traceability.

Req ID	Description	Status
SHRP-01	Reliable USB camera control (ISO, shutter, aperture, trigger)	Pending
SHRP-02	Real-time HFR calculation for star sharpness monitoring	Pending
SHRP-03	Remote focus stepping via USB	Pending
SHRP-04	Programmatic recovery from Sony USB hangs	Pending
CAPT-01	Aurora burst logic (adaptive exposure profiles)	Pending
CAPT-02	Histogram-based exposure guard rails	Pending
CAPT-03	Intervalometer with interrupted sequence support	Pending
MOB-01	Low-latency iPhone viewfinder from node preview	Pending
MOB-03	Telemetry dashboard (battery/temp/HFR/health)	Pending
SYNC-01	Multi-node discovery and synchronized triggering	Pending

Table 6. Initial v1 requirements traceability table (imported from project planning).

18 Astronomical Phenomena and Photographic Response

This section covers the physical phenomena that AuroraPhoto must handle—lunar phases, light-level dynamics, and the photographic trade-offs they impose. Understanding these phenomena motivates the adaptive control strategies in the preceding sections.

18.1 Lunar Eclipse Phases

During a lunar eclipse, the Moon passes through Earth’s shadow cones. The progression from penumbral entry (P1) through umbral contacts (U1–U4) to penumbral exit (P4) spans several hours, with the total phase (U2–U3) lasting up to ~107 minutes.

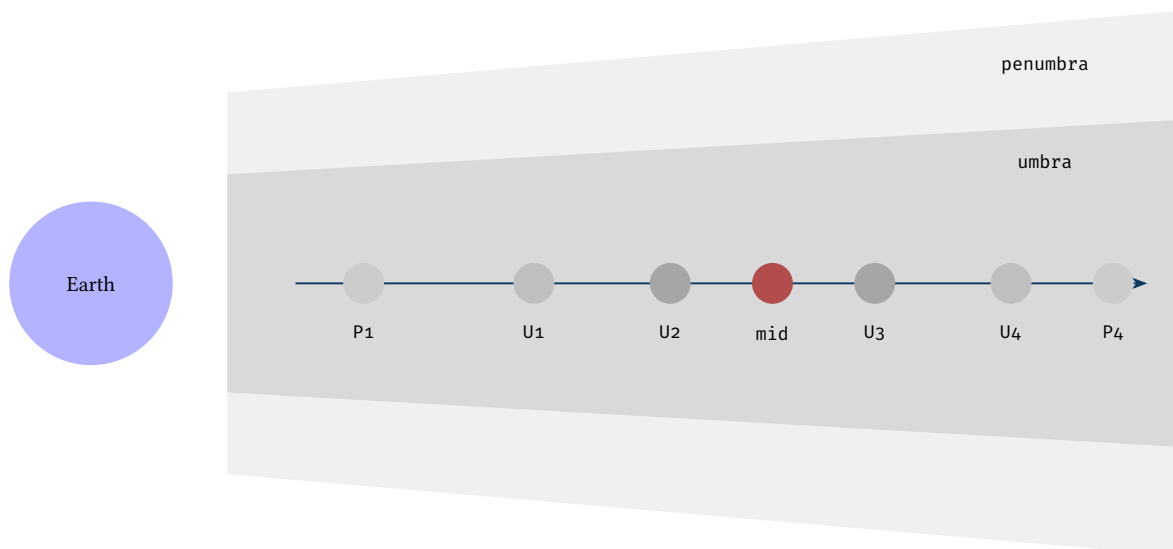


Figure 26. Earth shadow geometry: the Moon traverses the penumbral and umbral cones. P1/P4 mark penumbral entry/exit; U1–U4 mark umbral contacts; mid-eclipse occurs at deepest immersion.

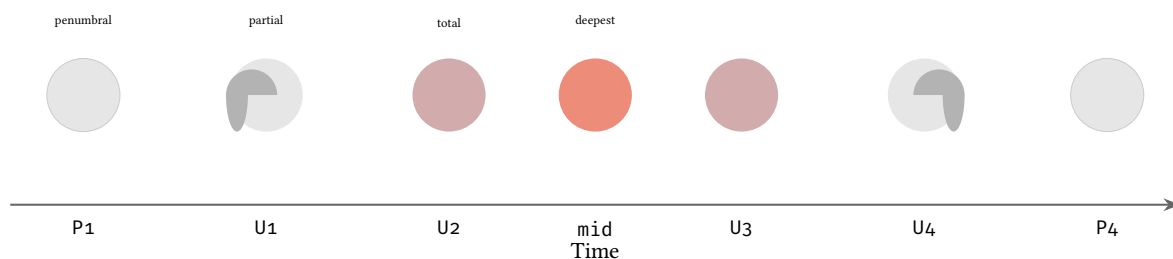


Figure 27. Lunar eclipse stage sequence: the Moon’s appearance progresses from subtle penumbral darkening through partial shadow entry to deep red totality and back.

18.2 Moon Phase and Night Sky Brightness

The Moon's phase is the dominant factor in night sky brightness for ground-based astrophotography. A full Moon raises the sky background by ~ 4 magnitudes per square arcsecond compared to a new Moon, dramatically reducing contrast for aurora and star imaging.

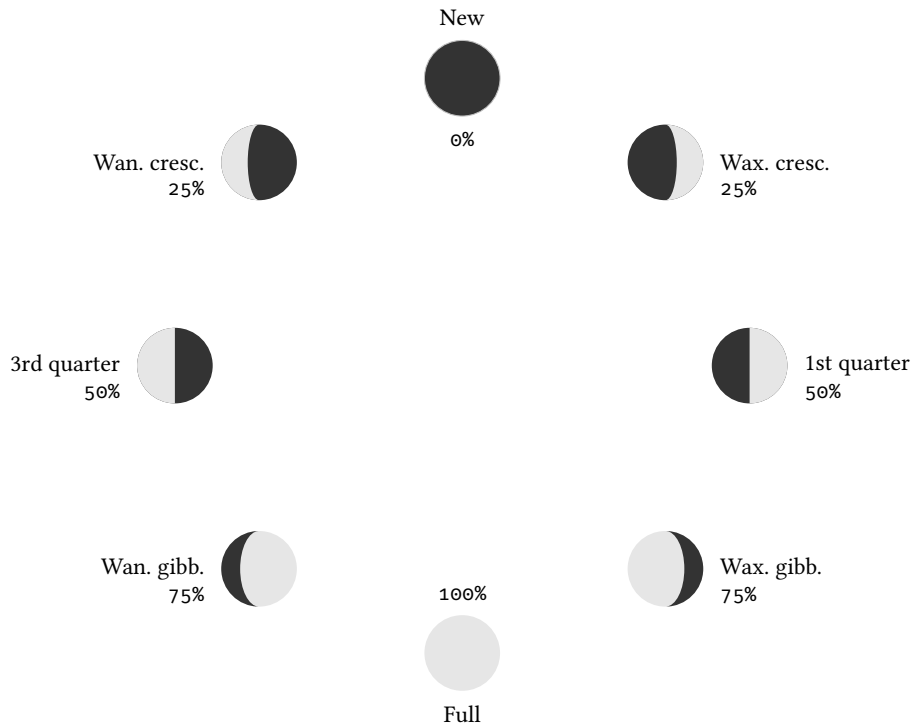


Figure 28. Lunar phase cycle with approximate illumination percentages. Each phase corresponds to a different sky brightness level and thus different imaging constraints.

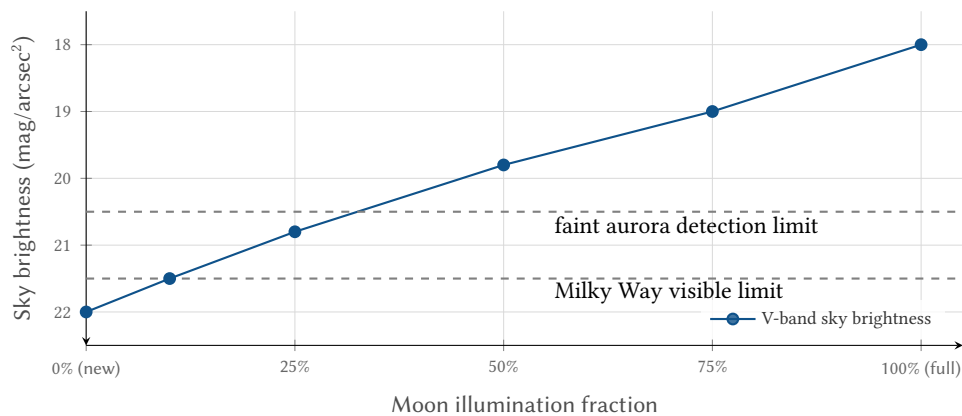


Figure 29. Sky brightness vs. moon illumination fraction. Lower magnitude values (top of plot) indicate brighter skies. Faint aurora and Milky Way structure become invisible as the sky brightens beyond their contrast limits.

Moon Phase	Sky (mag/arcsec ²)	Imaging Recommendation
New moon (<5%)	21.5–22.0	Ideal for all targets; faintest aurora and stars visible
Crescent (5–30%)	20.5–21.5	Good for aurora; mild sky glow, time shoots away from Moon
Quarter (40–60%)	19.5–20.5	Aurora still viable; stars washed in Moon-facing hemisphere
Gibbous (60–85%)	18.5–19.5	Bright aurora only; short exposures to manage sky glow
Full (>85%)	17.8–18.5	Difficult; only bright aurora (IBC III+) detectable

Table 7. Recommended imaging windows relative to moon phase. AuroraPhoto should factor moon phase into its exposure profile defaults.

18.3 Aperture Trade-offs for Astrophotography

Aperture selection in astrophotography involves balancing light gathering against optical aberrations. Wide apertures gather more light but introduce coma, vignetting, and field curvature; stopped-down apertures improve sharpness but reduce light and eventually hit the diffraction limit.

Wide open (f/2.8)	Stopped down (f/8)
Light: high (4× more)	Light: moderate
Diffraction: negligible	Diffraction: growing
Coma/aberrations: worst	Coma/aberrations: minimal
Depth of field: shallow	Depth of field: deep
Vignetting: significant	Vignetting: minimal
Star SNR: best	Star SNR: lower

Figure 30. Aperture effects comparison: wide open (f/2.8) maximizes light gathering and SNR at the cost of optical aberrations; stopping down (f/8) improves sharpness across the field but reduces light throughput.

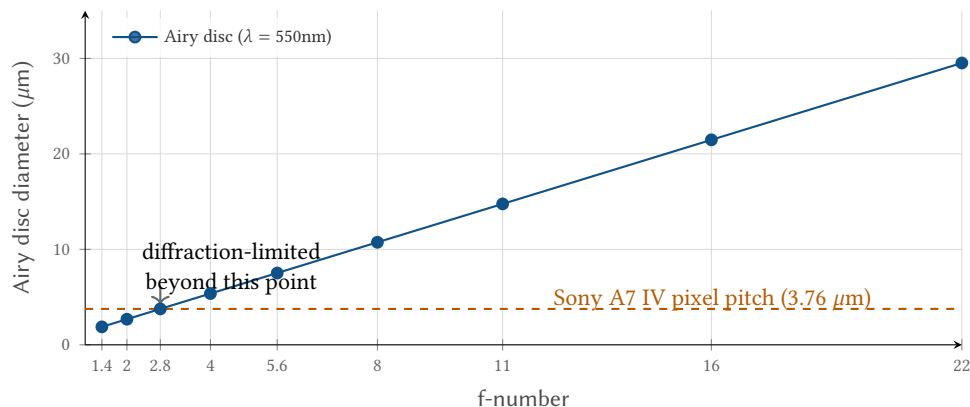


Figure 31. Diffraction-limited resolution vs. f-number: the Airy disc diameter grows linearly with f-number. Beyond the crossover with the sensor pixel pitch, further stopping down reduces resolving power.

Aperture	Pros	Cons
f/1.4–f/2	Maximum light; fastest shutter possible; best star SNR	Severe coma at edges; vignetting; narrow DoF; expensive glass
f/2.8–f/4	Good light/sharpness balance; reduced coma	Some vignetting; not diffraction-free on all lenses
f/5.6–f/8	Sharp across field; minimal vignetting; good for daytime solar imaging	Less light; longer exposures or higher ISO needed; approaching diffraction limit
f/11+	Maximum DoF; uniform illumination	Diffraction softening; significantly less light; not recommended for stars

Table 8. Aperture pros/cons matrix for astrophotography.

18.4 Shutter Speed Trade-offs and Motion Considerations

Shutter speed selection must account for three sources of apparent motion: Earth’s rotation (causing star trailing), the Moon’s orbital motion, and atmospheric seeing (high-frequency wavefront distortion). The classic “500 rule” and the more accurate NPF rule provide maximum exposure guidelines.

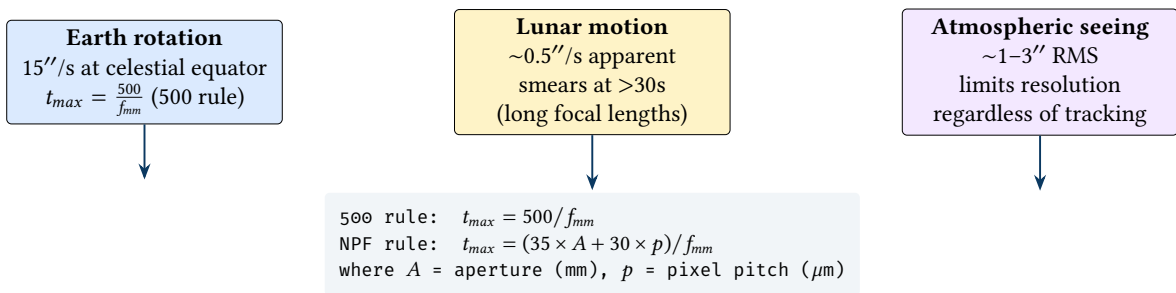


Figure 32. Three sources of motion blur in astrophotography, with formulas for maximum exposure before star trailing becomes visible.

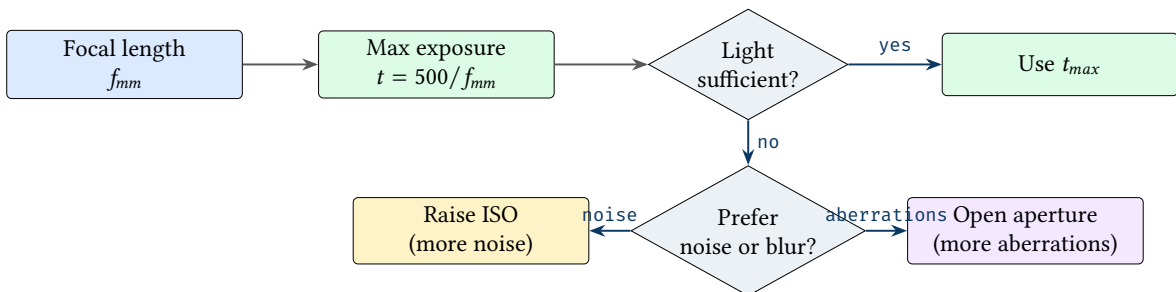


Figure 33. Shutter speed decision matrix: compute maximum exposure from focal length, then decide whether to raise ISO or open aperture if light is insufficient.

Shutter Speed	Pros	Cons
1/4000–1/1000s	Freezes fast transients; no trailing	Very little light; requires wide aperture or high ISO
1/500–1/60s	Good for solar/lunar imaging with filter	Still limited light for faint features
1/30–1/4s	Good aurora detail; captures faint neb- ulosity	Slight trailing at long focal lengths
1–4s	Faint aurora features; deep-sky tar- gets	Visible star trailing without tracking; higher noise from read noise accumulation
8–30s	Maximum light gathering for faint tar- gets	Pronounced trailing; requires equatorial mount; aurora smearing

Table 9. Shutter speed trade-offs with example scenarios for aurora and astrophotography.

18.5 Aurora-specific Light Adaptation

Aurora brightness is classified using the International Brightness Coefficient (IBC) scale, ranging from IBC I (barely visible, ~1 kR) to IBC IV (bright enough to read by, ~1000 kR). Aurora brightness is unpredictable and can change on timescales of seconds during substorm breakup events.

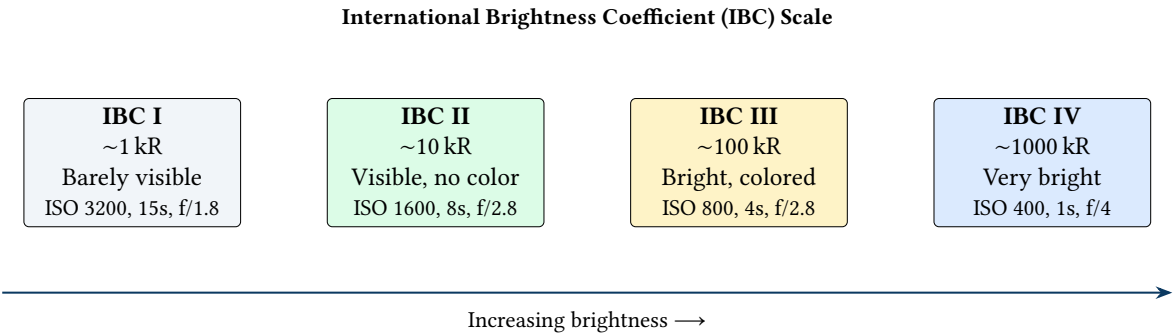


Figure 34. Aurora brightness classification (IBC I–IV) with corresponding camera settings. Each class requires different exposure parameters; AuroraPhoto’s burst detection maps to transitions between these classes.

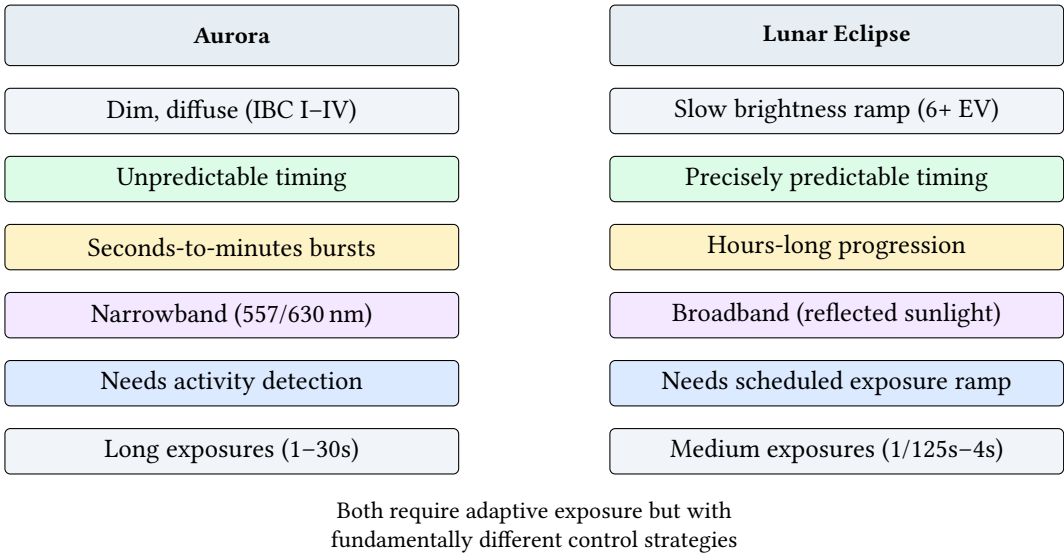


Figure 35. Aurora vs. lunar eclipse light adaptation comparison. Aurora requires reactive burst detection (see §3); lunar eclipses require scheduled exposure ramp sequences.

Cross-reference to System Architecture

The aurora IBC classifications map directly to the burst detection and exposure control loops described in Sections 3 and 10. IBC transitions correspond to the “activity score” change points that trigger profile switching in the burst loop. The lunar eclipse exposure ramp strategy maps to the automated parameter adjustment flowchart, providing a concrete use case for the histogram-based guard rails.

18.6 Spectral Characteristics and White Balance

Aurora and moonlight differ fundamentally in their spectral composition. Aurora emission is dominated by discrete atomic and molecular lines, while moonlight (reflected sunlight) produces broadband continuum light. These differences have direct implications for camera white balance, sensor channel weighting, and filter selection.

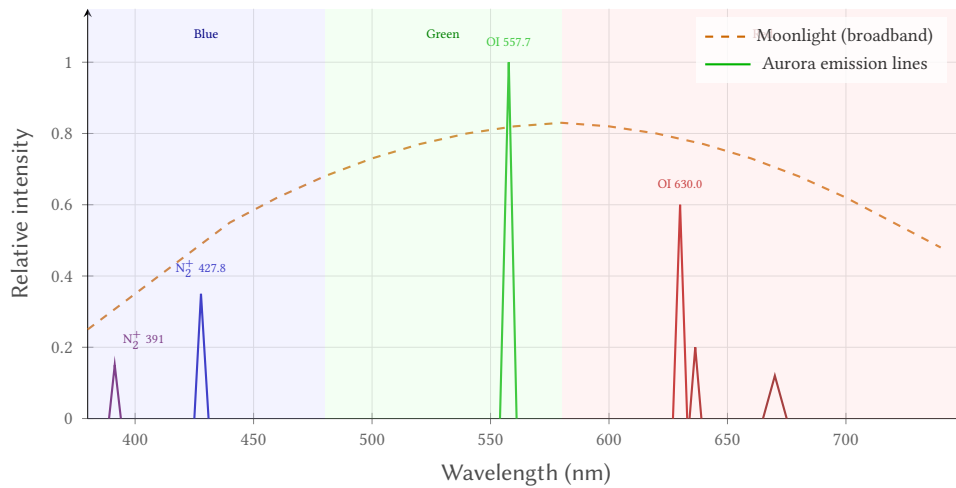


Figure 36. Spectral comparison: aurora produces discrete emission lines (dominant OI green at 557.7 nm, OI red at 630 nm, N_2^+ blue at 427.8 nm) while moonlight produces broadband continuum. Bayer filter channel boundaries shown as background shading. The green channel's sensitivity to the dominant 557.7 nm aurora line explains why green clips first.

18.7 Focal Length and Field of View Selection

Target selection drives focal length choice. Aurora curtains can span $10\text{--}90^\circ$, the lunar disc subtends $\sim 0.5^\circ$, and deep-sky objects vary widely. A field-of-view diagram helps match lenses to targets and informs the 500-rule calculations in the shutter speed subsection.

Insight:
The 557.7 nm green oxygen line falls squarely in the Bayer green channel, which has $2\times$ the pixel count of red or blue. This is why the green-channel clip avoidance rule (§3.3) is critical for aurora capture.

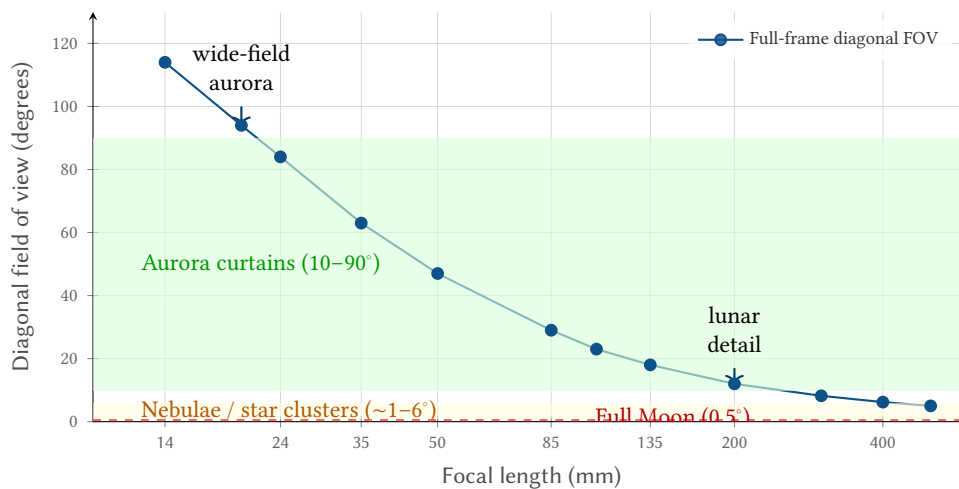


Figure 37. Focal length vs. diagonal field of view (full-frame sensor) with target angular sizes overlaid. Wide-angle lenses (14–35 mm) suit aurora and landscape context; telephoto (200–500 mm) is needed for lunar surface and deep-sky detail.

14–24 mm FOV: $84\text{--}114^\circ$ Aurora panoramas Milky Way	35–50 mm FOV: $47\text{--}63^\circ$ Aurora + landscape Star fields	85–135 mm FOV: $18\text{--}29^\circ$ Aurora detail Constellation scale	200–500 mm FOV: $5\text{--}12^\circ$ Lunar surface Deep-sky targets
500 rule: 20–35s	500 rule: 10–14s	500 rule: 3.7–5.9s	500 rule: 1.0–2.5s

Increasing focal length $\rightarrow$ narrower FOV, shorter max exposure

Figure 38. Lens focal length categories with target assignments and maximum untracked exposure times (500 rule). Longer focal lengths demand faster shutter speeds, compressing the available exposure range.

18.8 Optical Filters for Astrophotography

Filters serve different roles depending on the target: solar filters for solar and transit imaging, narrowband filters for isolating aurora emission, and light pollution filters for improving contrast in non-pristine skies.

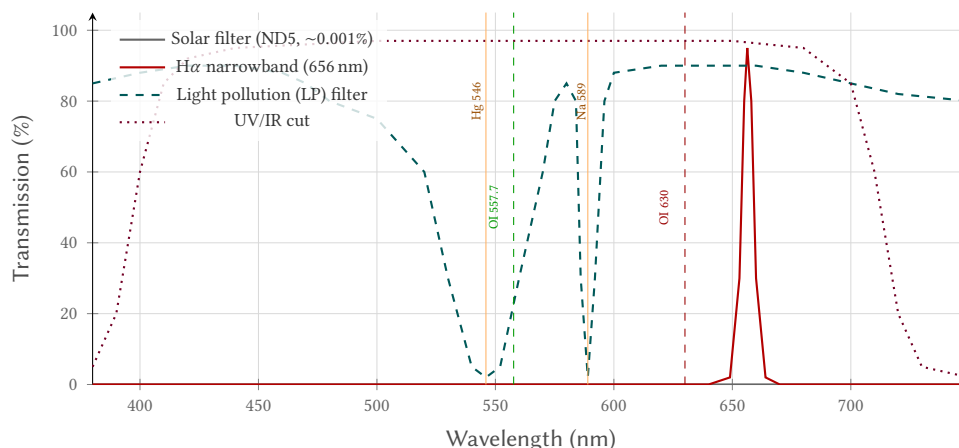
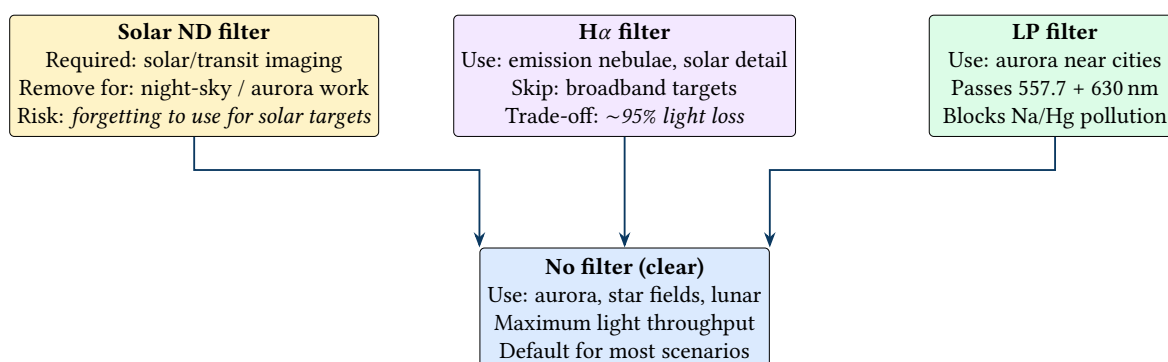


Figure 39. Filter transmission profiles: solar ND5 filters attenuate by ~5 orders of magnitude for safe partial-phase imaging; H α narrowband isolates chromospheric emission; light pollution filters reject sodium and mercury street lighting while passing aurora wavelengths.



Filter removal is a critical automated or operator action—
AuroraPhoto should track filter state and warn if inappropriate

Figure 40. Filter selection logic: each filter serves a specific target type. AuroraPhoto should track filter state and warn if a solar filter is absent during daytime solar imaging or still attached during night-sky work.

18.9 Atmospheric Effects on Imaging

Two atmospheric phenomena significantly impact astrophotography: extinction (absorption and scattering that increases with air mass at low elevation angles) and twilight gradients (the transition from sunlit to dark sky that defines the usable imaging window).

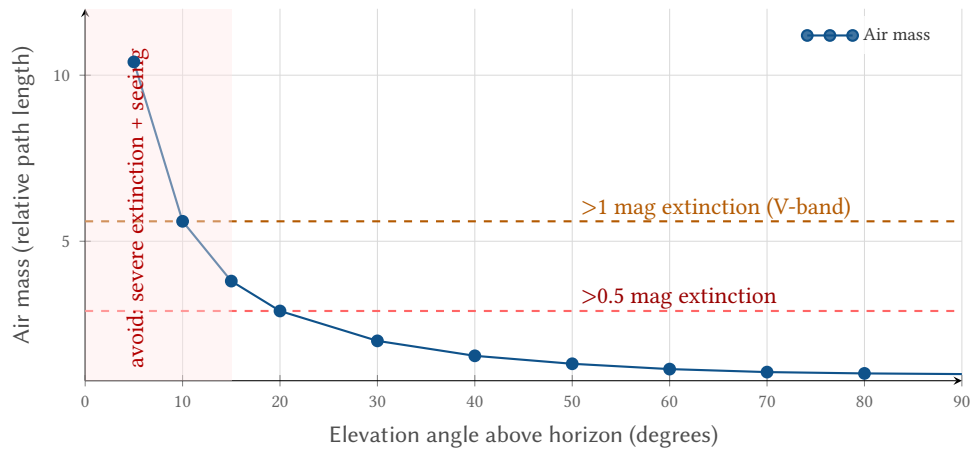


Figure 41. Air mass vs. elevation angle: atmospheric path length increases dramatically below 20° elevation, causing extinction (brightness loss), reddening, and degraded seeing. Targets below ~15° suffer severe quality penalties.

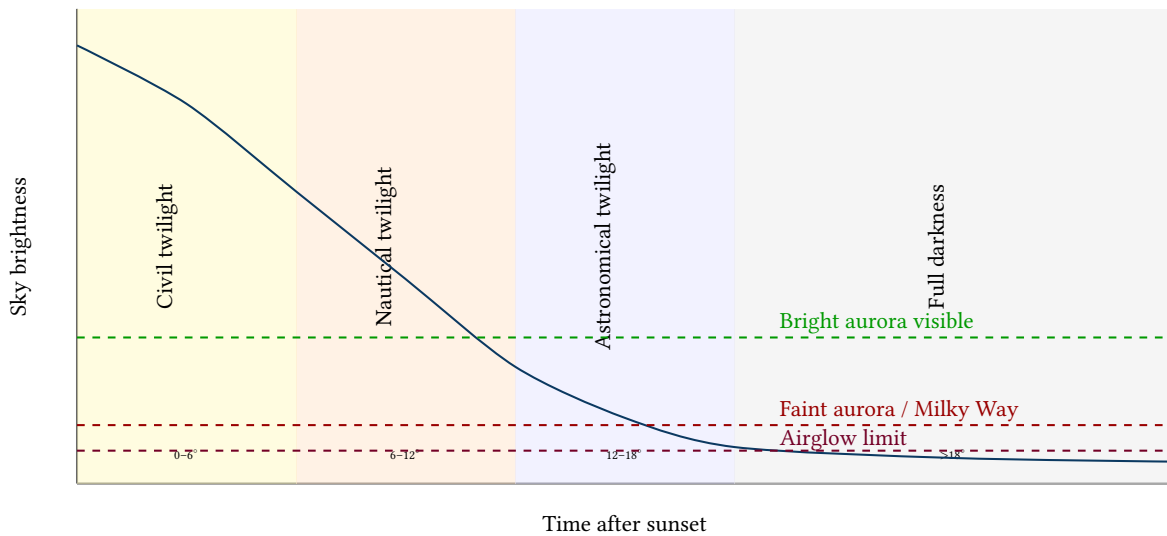


Figure 42. Twilight sky brightness progression: civil twilight (Sun 0–6° below horizon) is too bright for most astrophotography; nautical twilight allows bright aurora detection; full darkness (Sun >18° below) is needed for faint targets. AuroraPhoto should use solar depression angle to select initial exposure profiles.

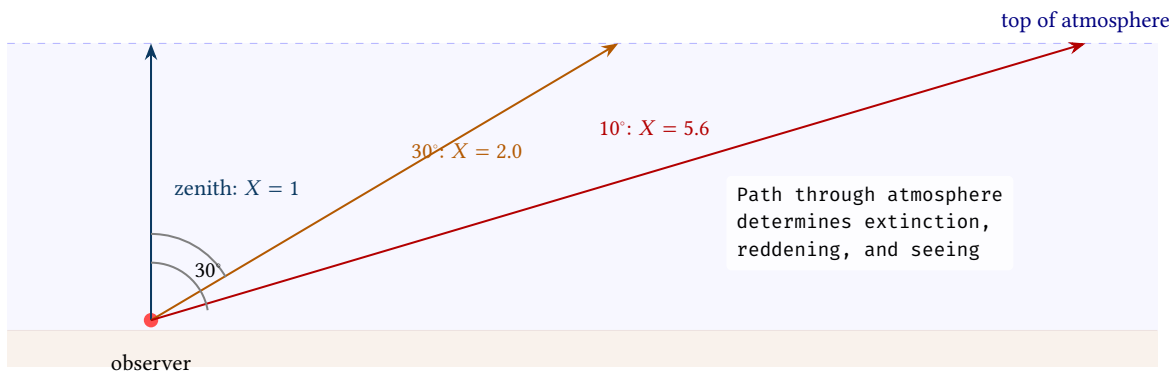


Figure 43. Air mass geometry: light from targets near the horizon traverses far more atmosphere than from the zenith, causing increased extinction (dimming), reddening, and scintillation.

18.10 Auroral Oval and Magnetic Latitude

Aurora occurrence is governed by Earth’s magnetic field geometry. The auroral oval—an annular region centered on the geomagnetic poles—defines where aurora is most frequently observed. Understanding this geography informs both deployment planning and the expected frequency of aurora events at a given observing site.

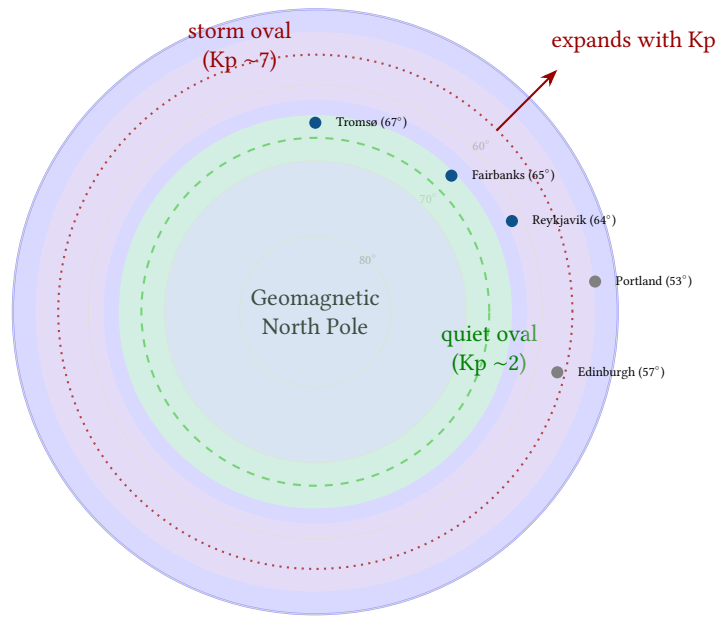


Figure 44. Auroral oval (polar view): during quiet conditions ($K_p \sim 2$) the oval sits at $\sim 65\text{--}70^\circ$ magnetic latitude; during geomagnetic storms ($K_p \sim 7+$) it expands equatorward, making aurora visible from mid-latitudes. AuroraPhoto deployments at sites within the quiet oval will see aurora most frequently.

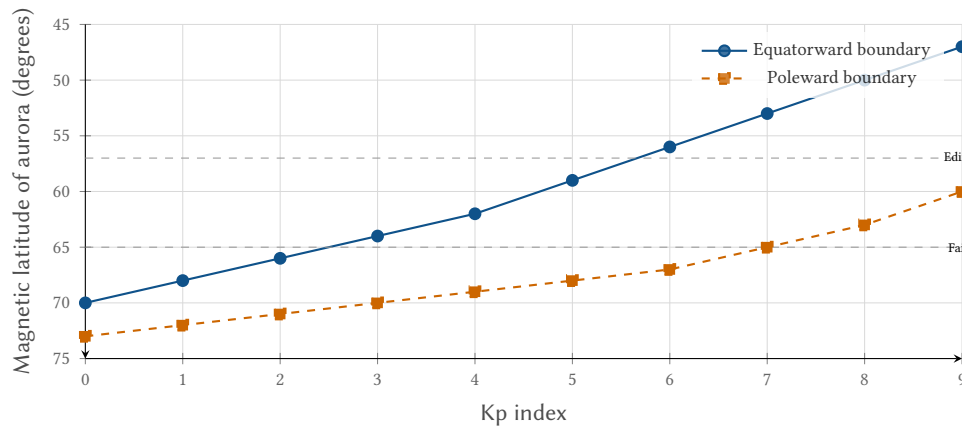


Figure 45. Auroral oval latitude vs. K_p index: higher geomagnetic activity pushes the oval equatorward. A site at 65° magnetic latitude sits under the oval for most K_p values; a site at 57° requires $K_p \geq 6$ for overhead aurora.

18.11 Solar Cycle and Aurora Frequency

The Sun follows an approximately 11-year activity cycle, driven by the reversal of its magnetic field. Solar maximum brings increased sunspot numbers, coronal mass ejections (CMEs), and geomagnetic storms—all of which drive aurora frequency and intensity. Planning multi-year imaging campaigns requires awareness of this cycle.

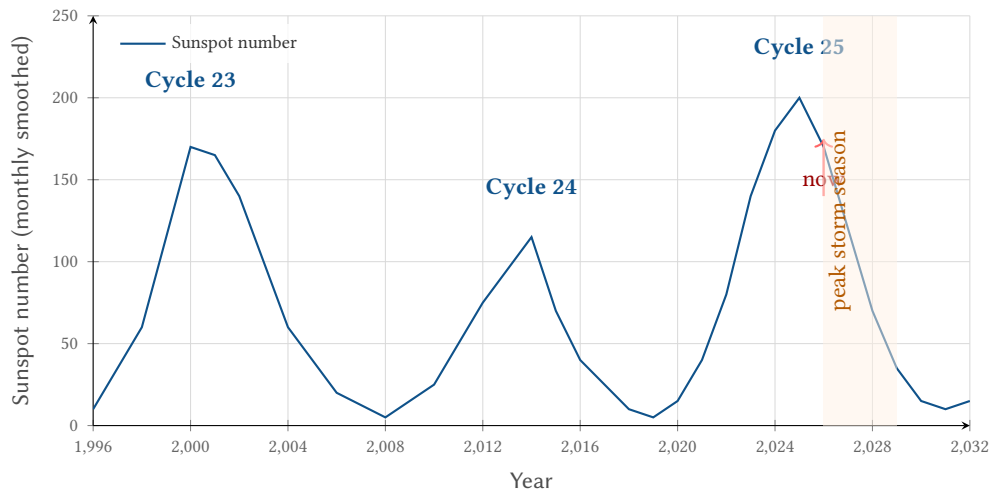
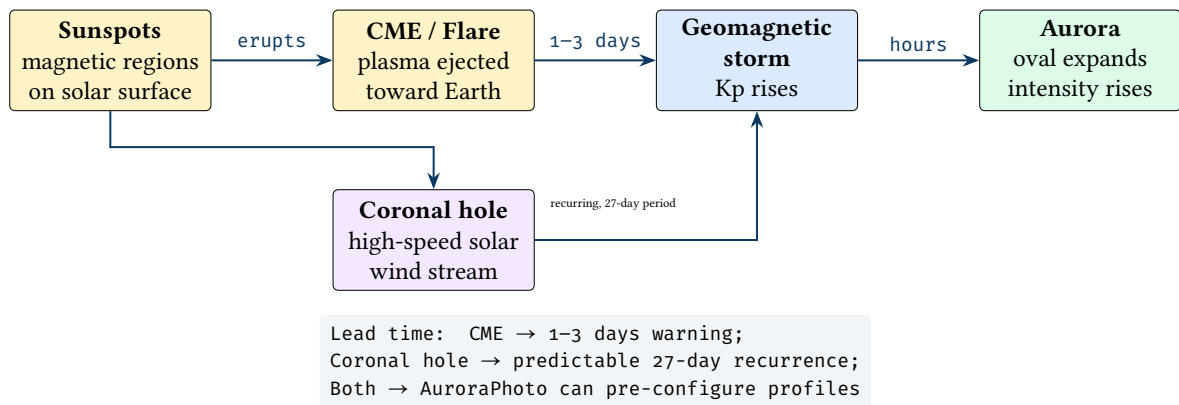


Figure 46. Solar cycle history and projection: Solar Cycle 25 is near its peak in 2025–2026, with the declining phase (2026–2029) historically producing the most intense geomagnetic storms and aurora. This is an optimal window for AuroraPhoto deployment.



Insight:
 The declining phase of the solar cycle (1–3 years after sunspot maximum) historically produces the most spectacular aurora due to coronal hole high-speed streams. The 2026–2029 window is prime time.

Figure 47. Solar-terrestrial chain: two pathways from solar activity to aurora. CMEs provide 1–3 days warning; coronal hole streams recur on a ~27-day solar rotation period, enabling advance planning.

18.12 Signal-to-Noise Fundamentals

Every imaging decision in AuroraPhoto ultimately affects signal-to-noise ratio (SNR). Understanding the noise model explains why the exposure ramp strategy (§13.5) prioritizes shutter speed over ISO, and why the histogram guard rails (§3.3) exist.

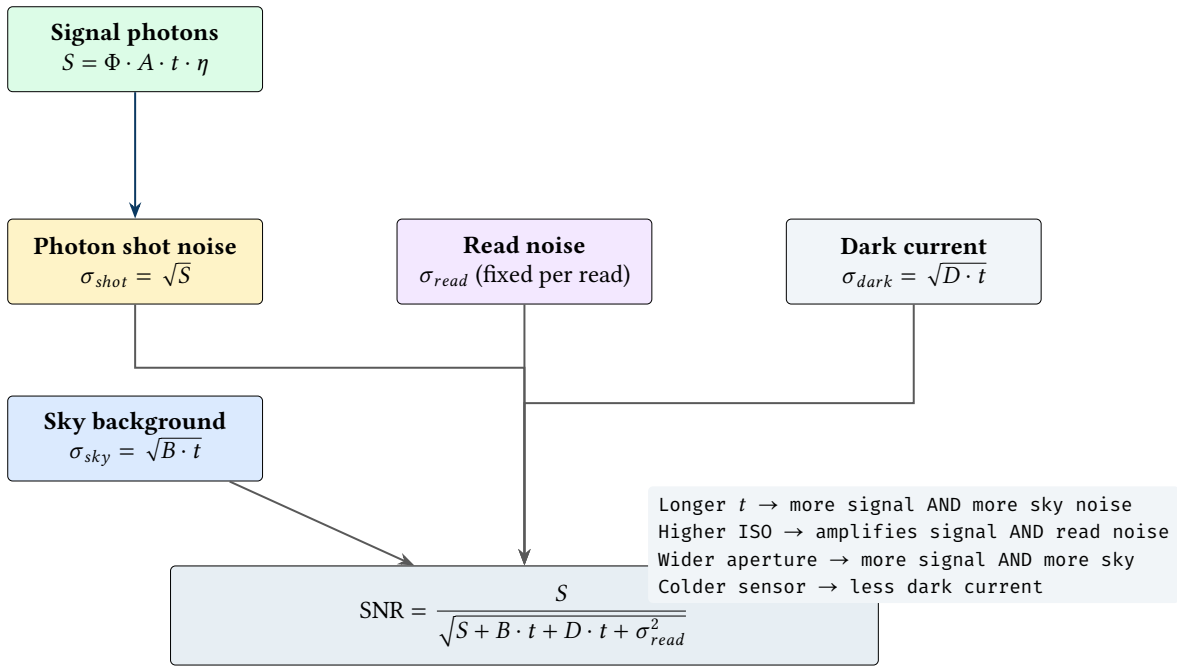


Figure 48. SNR model: signal is proportional to aperture area, exposure time, and quantum efficiency. Total noise combines photon shot noise, read noise, dark current, and sky background in quadrature. This model explains every exposure trade-off in the system.

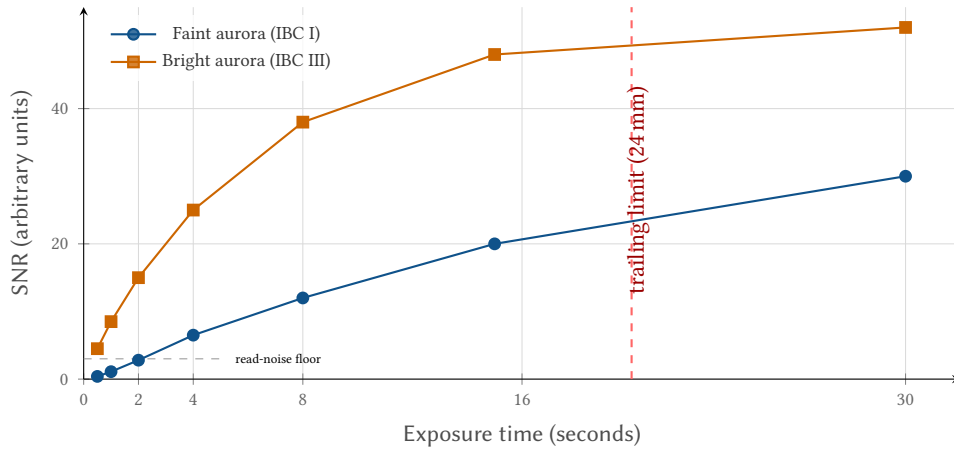


Figure 49. SNR vs. exposure time for faint and bright aurora targets. Faint targets are read-noise limited at short exposures and benefit greatly from longer integration; bright targets hit diminishing returns sooner. The star trailing limit imposes a hard ceiling on untracked exposure.

18.13 Histogram Interpretation for Astrophotography

The histogram is the primary feedback signal for AuroraPhoto's exposure guard rails. Different astronomical targets produce characteristic histogram shapes, and recognizing these patterns enables automated exposure decisions.

Insight:
 For faint aurora (IBC I–II), doubling exposure time from 8s to 16s increases SNR by ~40%. For bright aurora (IBC III+), the same doubling yields only ~20%. The control system should be more aggressive about long exposures for faint targets.

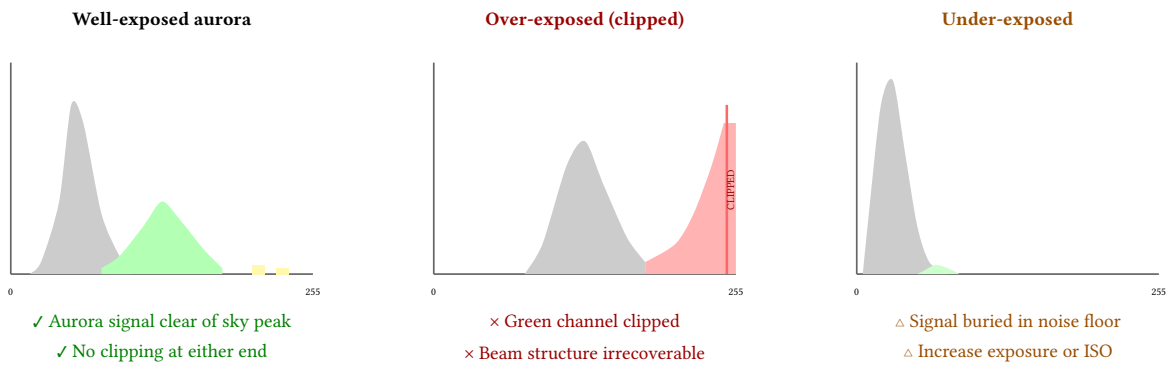


Figure 50. Histogram patterns for aurora imaging: well-exposed (left) shows aurora signal clearly separated from sky background with no clipping; over-exposed (center) has green channel piled against the right edge, destroying beam structure; under-exposed (right) buries the signal in the noise floor.

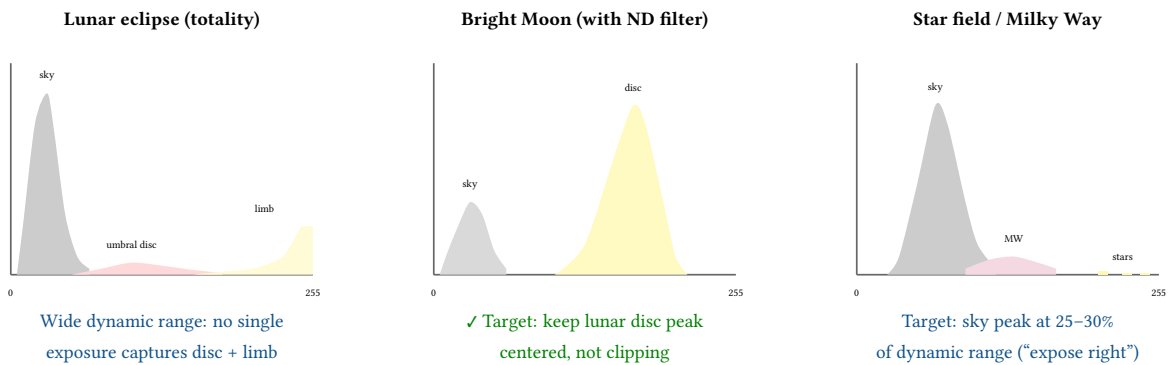


Figure 51. Histogram patterns for different targets: lunar eclipse totality (left) inherently exceeds single-exposure dynamic range; bright Moon with ND filter (center) should keep the disc peak centered; star fields (right) benefit from “exposing to the right” to maximize SNR while avoiding clipping.

: Histogram Guard Rail Thresholds

For aurora: trigger shutter reduction if any channel exceeds 95% of the histogram range. For star fields: target sky background peak at 25–30% of range (“expose to the right”). For lunar eclipse totality: clipping of individual exposures is expected and acceptable within a bracket sequence.

18.14 Star Point Spread Function and Focus Physics

The point spread function (PSF) of a star image results from the convolution of diffraction, optical aberrations, atmospheric seeing, and defocus. The Half-Flux Radius (HFR) used by AuroraPhoto’s focus loop (§3.2) is a single-number summary of the PSF width.

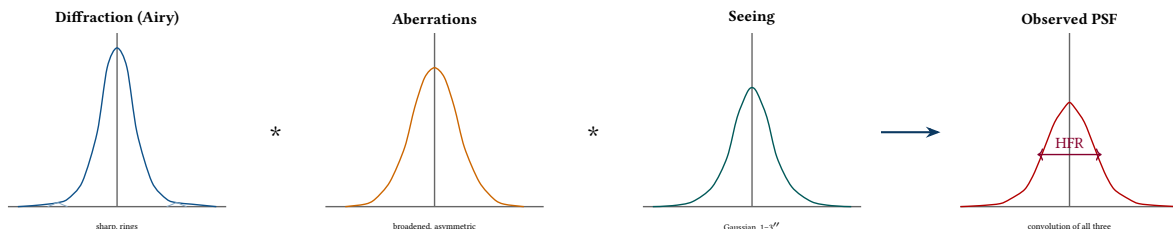


Figure 52. Star PSF composition: the observed profile is the convolution (*) of diffraction (Airy pattern), optical aberrations (lens-dependent), and atmospheric seeing (Gaussian). HFR measures the half-light radius of this combined profile.

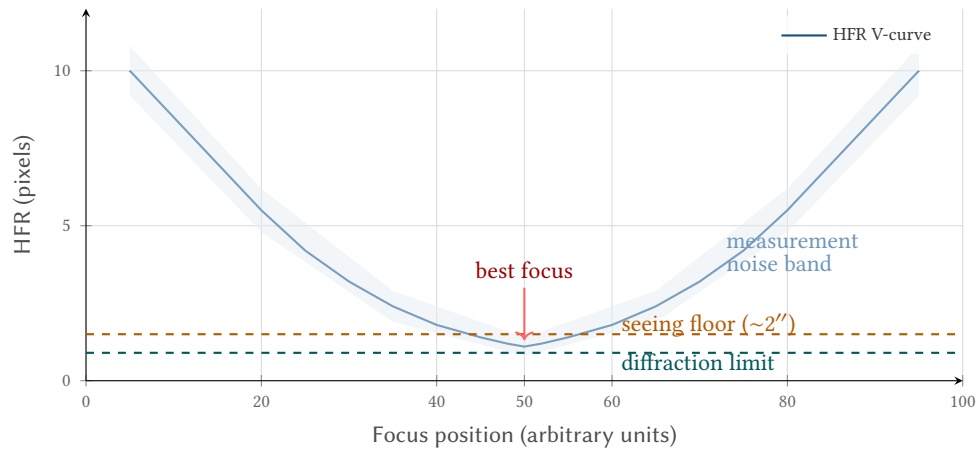


Figure 53. Focus V-curve: HFR follows a characteristic V-shape with minimum at best focus. The shaded band represents measurement noise (from photon statistics and sky background). The diffraction limit and seeing floor impose a hard minimum that even perfect focus cannot surpass.

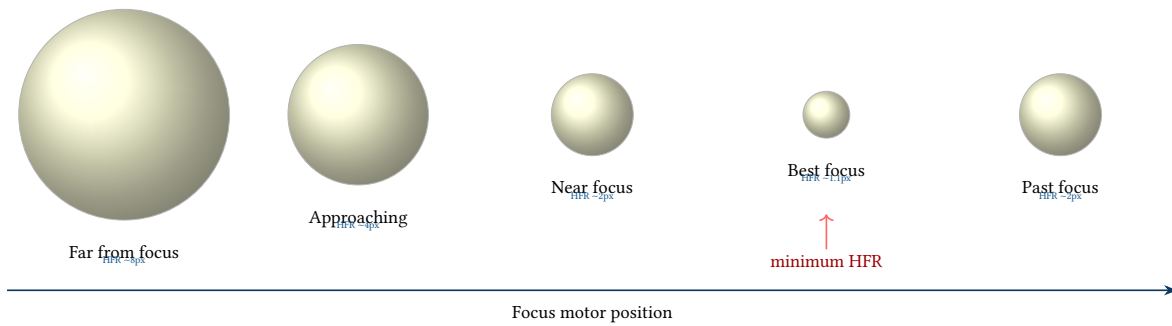


Figure 54. Visual representation of star images at different focus positions: defocused stars appear as large discs; the focus loop minimizes disc size (HFR) by stepping the motor to the V-curve minimum.

18.15 Multi-camera Deployment Geometry

An AuroraPhoto deployment typically uses multiple camera nodes for complementary coverage. The optimal geometry depends on the target: parallax-based 3D reconstruction requires baseline separation, panorama stitching requires overlapping fields, and multi-focal-length setups require co-located but differently configured nodes.

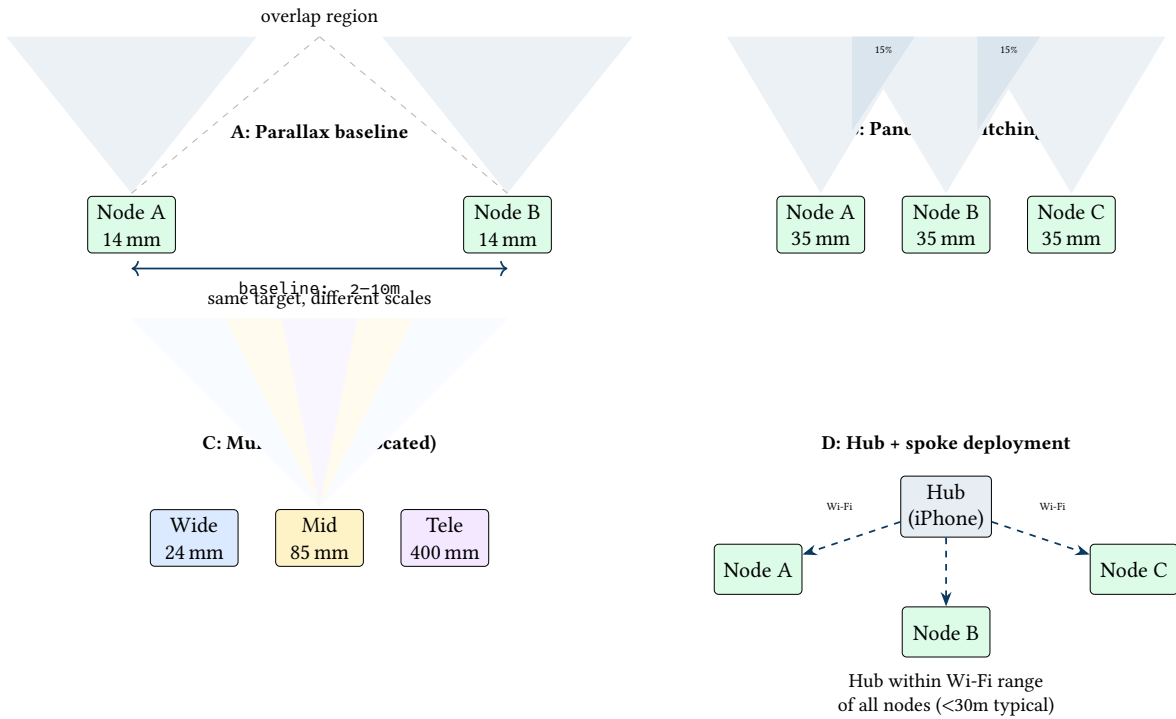


Figure 55. Multi-camera deployment geometries: (A) parallax baseline for potential 3D reconstruction, (B) panorama stitching with ~15% overlap, (C) co-located multi-focal-length for simultaneous wide/narrow coverage, (D) hub-and-spoke Wi-Fi topology constraint.

18.16 Image Stacking for Noise Reduction

Averaging N aligned frames of the same scene improves SNR by $\sqrt{N}$, enabling faint aurora and star detail to emerge from the noise floor. For untracked aurora frames, alignment must handle both the moving aurora and the (slowly) trailing stars separately.

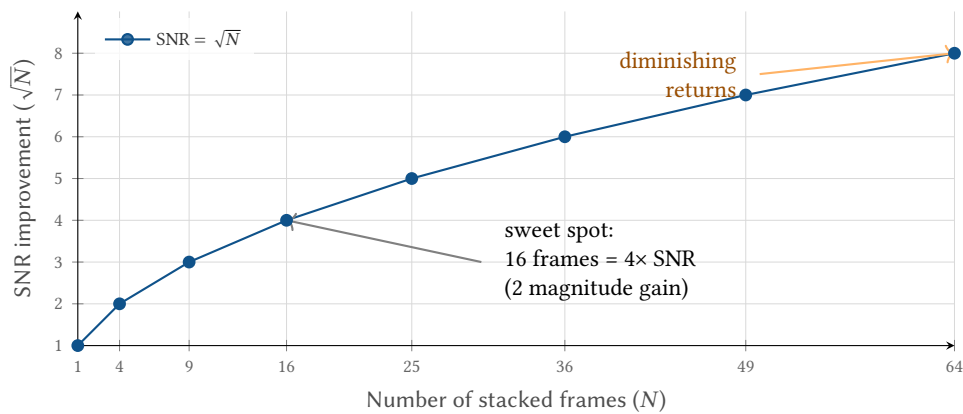


Figure 56. SNR improvement from stacking: 16 frames yield a 4× SNR improvement (equivalent to 2 magnitude gain), revealing structure invisible in single exposures. Beyond ~25–36 frames, returns diminish and other error sources (alignment, sky variation) dominate.

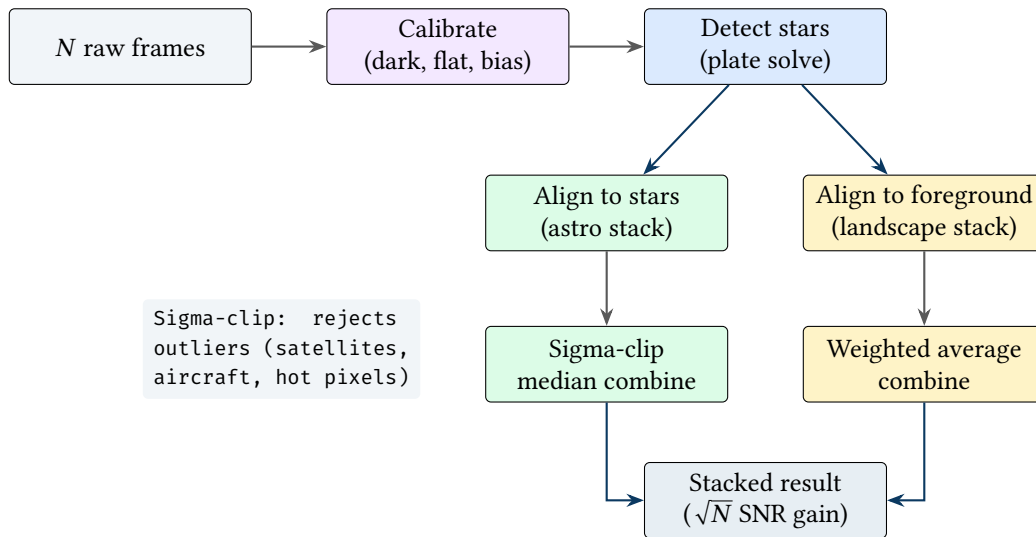


Figure 57. Image stacking pipeline: raw frames are calibrated, stars are detected for alignment, and frames are combined using either star-aligned stacking (sharp stars, blurred foreground) or foreground-aligned stacking (sharp landscape, trailed stars). Sigma-clipping during combination rejects transient artifacts.

18.17 Sky Quality and the Bortle Scale

The Bortle Dark-Sky Scale classifies observing sites from pristine (Class 1) to inner-city (Class 9). Sky quality directly determines the faintest aurora detectable and the exposure settings needed. AuroraPhoto should account for site Bortle class when selecting default exposure profiles.

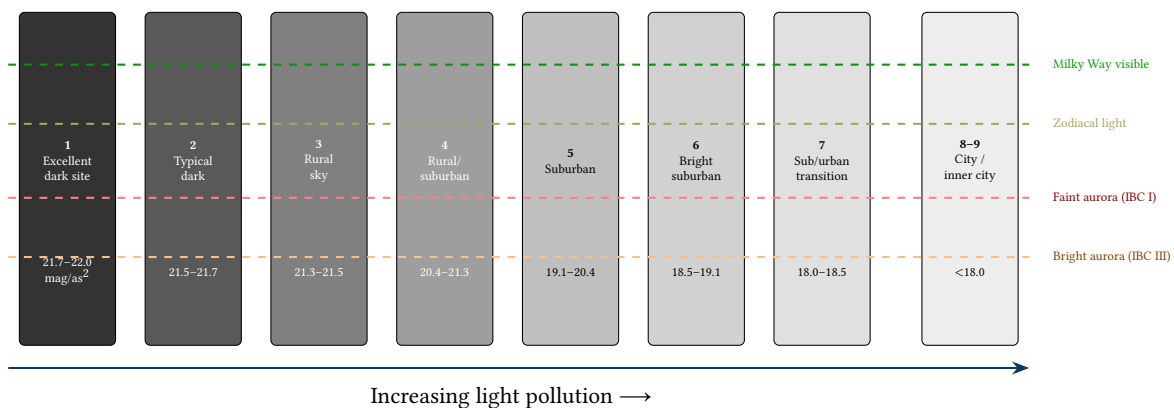


Figure 58. Bortle Dark-Sky Scale: site quality determines the faintest targets detectable. Faint aurora (IBC I) requires Bortle 4 or better; bright aurora (IBC III+) is visible even from Bortle 6–7 suburban sites. AuroraPhoto should factor Bortle class into default exposure profile selection.

18.18 Weather and Clear Sky Planning

Cloud cover is the single largest risk factor for astronomical photography—all other planning is moot under overcast skies. Statistical clear-sky probability varies dramatically by location and season, and real-time satellite cloud imagery enables go/no-go decisions.

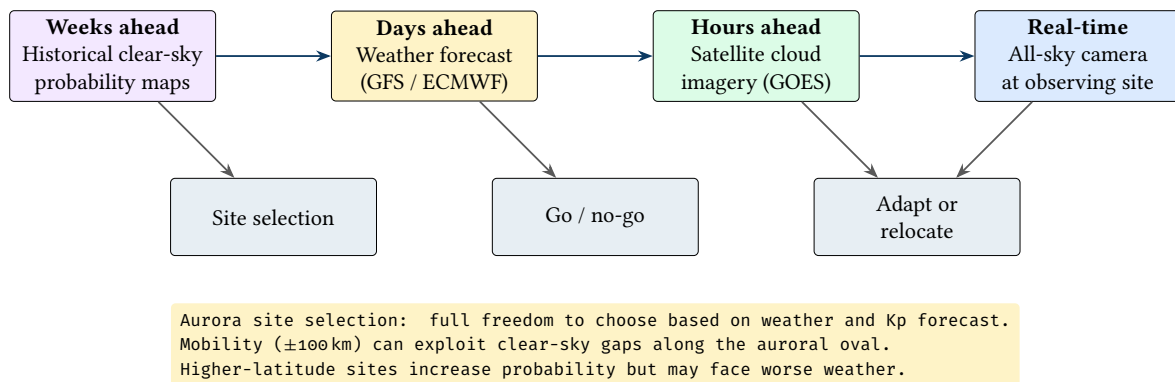


Figure 59. Weather planning cascade: long-range climatology informs site selection, medium-range forecasts drive go/no-go decisions, and real-time satellite imagery enables last-minute adaptation. Aurora observers have full site-selection freedom within the auroral oval.

Revision History

Version	Date	Changes
0.1.0	March 2026	Initial technical report skeleton; architecture + control-loop diagrams